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Robust real-time strawberry maturity detection using UAV-mounted deep learning for precision agriculture

Rajmeet Singh¹, Appaso M. Gadade² and Irfan Hussain^{1*}

Abstract

Background To address the challenge of real-time plant monitoring in greenhouse environments, this industry-driven research focuses on developing an autonomous quadrotor UAV system specifically designed for monitoring strawberry plants. Traditional methods for greenhouse monitoring are labor-intensive and lack scalability, particularly in precision agriculture applications.

Method This research introduces a mature strawberry detection model specifically designed for greenhouse environments. The proposed YOLOv9-GLEAN approach enables the identification of small mature strawberries through an onboard camera mounted on the quadrotor. Additionally, a hybrid trajectory tracking controller for the quadrotor is developed and tested in both simulated and real-world conditions. The UAV navigates through the greenhouse using predetermined waypoints, operating as a semi-autonomous system for navigation while maintaining full autonomy in mature strawberry detection tasks. The system incorporates an integrated onboard vision platform that utilizes an innovative YOLOv9-GLEAN-based algorithm to perform real-time and offline detection and counting of mature strawberries.

Results The YOLOv9-GLEAN model achieves high detection accuracy, as confirmed by evaluation metrics such as precision, recall, and F1-score. The proposed hybrid (PID+LQR) controller demonstrates superior tracking performance compared to other conventional controllers. The integrated control and perception system proves effective in both simulated and real-world greenhouse environments.

Discussion The research validates the efficacy of deep learning models, with YOLOv9-GLEAN showing exceptional performance in enabling rapid, precise, and automated detection of ripe strawberries through quadrotor deployment in greenhouse environments. Such agricultural monitoring technologies represent a substantial advancement beyond conventional manual inspection approaches, empowering farmers and greenhouse operators to execute well-informed, time-sensitive management decisions that minimize crop losses and optimize production yields. This investigation underscores the revolutionary impact that deep learning technologies can have within greenhouse agriculture.

Keywords UAV, Computer vision, Trajectory tracking, Strawberry fruit, Deep learning, Remote sensing

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Introduction

Greenhouse cultivation has become a crucial solution for tackling the challenges of modern agriculture, including limited cultivated land, unpredictable climatic conditions, and the increasing demand for high-quality crop production. Monitoring in greenhouses is essential for multiple reasons. It facilitates the early detection of stress caused by diseases, pests, or unfavorable environmental conditions, enabling prompt interventions to reduce yield losses. Additionally, it aids in identifying the optimal time for harvesting fruits, ensuring better quality and productivity [56, 61].

The accelerated advancement of artificial intelligence in recent years has driven significant progress in deep learning technologies. Deep learning enables machines to emulate human cognitive processes including vision, and classification to address sophisticated pattern recognition problems, leading to successful applications in rapid agricultural crop and fruit identification.

Traditional computer vision approaches for fruit detection in agricultural settings face several significant limitations. Classical methods such as template matching, feature-based detection (SIFT, SURF), and color-based segmentation techniques struggle with the inherent variability present in natural environments [1]. These conventional approaches are particularly challenged by varying lighting conditions, occlusion from leaves and branches, diverse fruit orientations, and the complex background textures commonly found in greenhouse settings [52]. Furthermore, traditional methods often require extensive manual feature engineering and threshold tuning for different environmental conditions, making them impractical for real-world deployment where adaptability and robustness are essential [30]. Additionally, these approaches typically exhibit slow processing speeds when dealing with high-resolution imagery, making real-time applications computationally prohibitive [1].

Given these limitations, deep learning-based object detection frameworks, particularly the You Only Look Once (YOLO) architecture, present compelling advantages for agricultural monitoring applications [45]. YOLO was specifically selected for this study due to its unique single-stage detection approach that simultaneously predicts bounding boxes and class probabilities, enabling real-time processing capabilities essential for UAV-based monitoring [62]. Unlike two-stage detectors such as R-CNN and Fast R-CNN that require separate region proposal and classification steps, YOLO's unified architecture provides the optimal balance between detection accuracy and computational efficiency required for deployment on resource-constrained UAV platforms. Moreover, YOLO's end-to-end learning capability eliminates the need for manual feature engineering while demonstrating superior generalization across varying

environmental conditions commonly encountered in greenhouse agriculture.

Cape gooseberry fruit maturity was detected [11]. through integrated machine learning methodologies combined with multiple color space analyses technique. Xu et al. [69] implemented the YOLOv3 detection framework for identifying various developmental stages and classifications of tea bud varieties. Strawberry identification methodologies within greenhouse settings were investigated by Harshana et al. using deep convolutional neural networks, with mean detection accuracy of 77.21% being achieved [24]. Chen et al. expanded the detection scope to include strawberry flowers alongside ripe and unripe fruit classification, implementing yield estimation through the FAST-RCNN algorithm with average detection accuracy of 72% [13]. Li et al. employed combined depth feature extraction and classifier techniques for strawberry appearance recognition, primarily targeting mechanized production systems with accuracy reaching 96.55%, though the method's reliance on uniform backgrounds limited its applicability in natural growing conditions [75].

Drone-based solutions for crop monitoring are widely popular and highly reliable in the agricultural sector, due to their advanced computing capabilities and onboard sensors [26, 47]. In recent years, unmanned aerial vehicles (UAVs) have been extensively utilized in the agricultural sector [16]. The multi-spectral camera is mounted on the hexacopter to monitor the plant growth through ortho-images of the plants [7]. UAV based citrus trees disease detection (Huanglongbing) was proposed in [19]. The drones are used to detect the weeds in the open field crops using deep learning approach [46]. UAVs are utilized for monitoring and managing diseases in strawberry crops [13, 33, 49, 77].

This study focuses on the design and simulation of a flight controller for UAVs, specifically tailored for plant monitoring within greenhouse environment. The Flight Control System (FCS) serves as the brain of a quadrotor UAV, with the responsibility of maintaining the desired attitude and position during flight. Factors such as wind disturbances, coupled nonlinear dynamics, and modeling uncertainties pose significant challenges for control researchers in developing effective flight control systems [38]. Adaptive control is well-suited for addressing uncertainties arising from the operating environment [44]. Model reference adaptive control for quadrotor to tackle uncertainties and external disturbance is proposed in [34]. The Proportional-Integral-Derivative (PID) controller is the most commonly used control method in UAVs for real-time applications. It is a linear controller known for its simplicity and ease of tuning [42]. Romero et al. linearized the quadrotor model and proposed a PID controller for stabilization [50]. The authors showed that

the PID controller is effective only within a range of ± 10 degree for roll and pitch angles, and its performance is significantly influenced by external disturbances. The use of a PID controller for a quadrotor presents several limitations. It is primarily suitable for linear systems, its control performance is not good for wider roll and pitch angles, and it lacks robustness against external disturbances. In [55], Linear Quadratic Regulator (LQR) controller outperforms the PID controller in quadrotor systems and is applied to the linearized model of the UAV. However, the LQR controller is sensor-intensive [12]. Padhy et al. proposed a Deep Neural Network (DNN) for UAVs [41].

Currently, no research has specifically explored the use of UAVs for strawberry (*Fragaria* spp.) plant monitoring in greenhouse environments for simulation and real-world scenario. This paper aims to address this gap by proposing both a simulation and real time based investigation of a quadrotor flight controller designed for precise navigation within greenhouse settings and detecting mature strawberries. The plant monitoring tasks include fruit maturity classification, disease detection, yield estimation, and growth assessment. The primary focus of this research is to develop a deep learning-based model for detecting mature strawberries, along with a hybrid flight controller for a quadrotor within greenhouse. The system is designed to effectively detect and monitor strawberry plants while accurately following a predefined trajectory within a greenhouse environment. The novelty of the work is that the quadrotor follows the predefined waypoints for monitoring the strawberry plants when GPS is not working in the greenhouse due to Polyethylene (PE) Film roof sheets. Hence, the aim of this study is to design and develop a quadrotor hybrid controller with capability to detect mature strawberries for plant monitoring in the greenhouse settings with the following major contributions:

- Proposed a novel deep learning based YOLOv9-GLEAN detection model tailored for strawberry detection from quadrotor-captured videos in both online and offline mode. Also presented state-of-the-art comparison.
- Developed and validated a quadrotor hybrid trajectory tracking controller for both simulated and real greenhouse environment.
- Created a digital twin model of strawberry greenhouse environment for accurate replication of operational conditions and high-precision waypoint-based trajectory tracking.
- The developed system demonstrates end-to-end capabilities by performing real-time detection and counting of ripe strawberries using quadrotor onboard vision sensors. The approach was validated

through extensive experimentation in both simulated environments and real greenhouse conditions, highlighting its robustness and applicability.

To comprehensively explore the topic, this manuscript adheres to the following structure: “**Related work**” section: represents the related literature on the detection of crop, providing a comprehensive overview. “**Methodology**” section introduces a methodology for accurately detecting mature strawberries using deep learning technique and modeling of quadrotor and design and development of quadrotor trajectory tracking controller. “**Experiments**” section covered the technical details about the implementation and simulation framework and provides the detailed experimental results of the work in both simulation and real experimentation. “**Discussion**” section, presents the discussion, while “**Conclusion**” section summarizes the insights, contributions of this study and future scope.

Related work

Recent research in deep learning has focused on developing more powerful system architectures and learning methods, such as CNNs, Transformers, Perceivers, and Mambas [67, 68, 78]. The You Only Look Once (YOLO) series [5, 20, 63] has emerged as a leading approach in real-time object detection. These models commonly rely on core computational units such as Cross Stage Partial Network (CSPNet) [64], Efficient Layer Aggregation Network (ELAN) [65], or their respective variants. A comprehensive methodology for labeling, training, and detecting tomato flowers tailored for robotic pollination was proposed by Singh et al [60]. Transfer learning techniques were employed using YOLOv5 and YOLOv8 models for the detection of tomato flowers in the context of pollination. A multi-level deep information feature fusion extraction network (DFN-PSAN) was proposed by Dai et al. [15] for accurate plant disease classification in natural field environments. The DFN-PSAN model was built upon the YOLOv5 Backbone and Neck network as its base structure (DFN), and a pyramidal squeezed attention (PSA) mechanism was integrated to enhance classification performance. An improved YOLOv5-based model was proposed for detecting and grading sprouted potatoes in complex scenarios [14]. To enhance feature representation and mitigate the issue of feature similarity loss during the fusion process, the Conv layers in the C3 module were replaced with CrossConv. Recently, real-time detectors such as RT-DETR [74], based on the DETR framework [10], have been introduced. However, adaptation of the DETR series to new domains has been found to be challenging without the use of domain-specific pre-trained models. As a result, the YOLO series has

been widely adopted as the most commonly used real-time object detector.

CNN-based models

CNN-based object detection frameworks have been widely utilized in precision agriculture for identifying and monitoring fruit crops such as strawberries. Early studies applied standard Convolutional Neural Networks (CNNs) like VGGNet and AlexNet for fruit classification tasks, demonstrating the effectiveness of deep feature extraction in distinguishing strawberries from background clutter [58, 73]. To achieve better localization, Region-based CNN (R-CNN) and its successors—Fast R-CNN and Faster R-CNN—were explored for detecting strawberries in both open-field and greenhouse settings. These two-stage detectors offered high detection accuracy by combining region proposal mechanisms with CNN feature extraction, although their inference speed was limited for real-time deployment [21, 48].

Further improvements were made through the integration of Single Shot MultiBox Detector (SSD), which simplified the detection pipeline by eliminating the need for region proposals, enabling faster processing while maintaining reasonable accuracy in moderately cluttered scenes [32]. Studies applying SSD for strawberry detection have shown that it performs well under conditions with moderate occlusion and varying illumination, often encountered in greenhouse environments. Moreover, hybrid approaches integrating pre-trained CNN backbones (e.g., VGG16 or ResNet) with custom detection heads have also been explored to enhance feature generalization across diverse growing conditions [52, 72]. Overall, CNN-based detectors, especially those using R-CNN and SSD architectures, remain a strong foundation for automated strawberry plant monitoring, particularly in controlled environments where real-time speed is not a strict constraint.

YOLO and transformer-based models

Accurate and real-time detection of strawberries in complex agricultural environments has been widely addressed using deep learning, particularly through CNN-based architectures like the YOLO series and more recently, transformer-based models. The YOLO (You Only Look Once) family of models has been extensively applied due to its speed and efficiency. Early versions like YOLOv3 and YOLOv4 were successfully used to detect ripe strawberries in field and greenhouse conditions, demonstrating robustness against partial occlusion and variable lighting [57]. Subsequent models such as YOLOv5 and YOLOv8 introduced architectural enhancements including CSPNet and PANet, leading to improved detection accuracy and reduced inference time. These models have been deployed on robotic platforms for tasks such as fruit

counting, flower detection, and autonomous harvesting [2, 3, 8, 60].

In parallel, transformer-based object detectors have emerged as a powerful alternative, offering superior capability in modeling global relationships across an image. DETR (DEtection TRansformer) introduced a fully attention-based detection framework that eliminates the need for region proposals or anchor boxes [10]. However, due to limitations in convergence speed, improved variants such as Deformable DETR and Conditional DETR have been adopted in agricultural settings. For instance, Zheng et al. [76] used a Swin Transformer backbone for multi-scale strawberry detection in greenhouses, outperforming traditional CNNs in densely packed scenes. Similarly, Lu et al. [31] integrated Vision Transformers (ViT) into a multimodal framework that combined RGB and depth data for precise fruit localization.

Overall, YOLO-based detectors remain the preferred choice for real-time applications due to their lightweight architecture, while transformer-based models offer improved accuracy in complex scenarios and are gaining traction in precision agriculture.

This study introduces an efficient deep learning-based framework for real-time strawberry identification, leveraging recent advancements in computer vision and smart agriculture. The proposed system, named Mature Strawberry Detection (MSD), incorporates the YOLOv9-GLAN model to achieve high-speed and accurate object detection. The trained MSD model was deployed on a quadrotor equipped with a camera, aiming to bridge the gap between controlled laboratory research and practical field applications. This configuration enables automated, contactless, and real-time monitoring of strawberry plants within a greenhouse environment. The use of a quadrotor enhances flexibility and precision in scanning mature strawberry plants, facilitating accurate detection and timely harvesting. By combining deep learning with robotic automation, the MSD framework provides a scalable and effective solution for modern greenhouses, reducing reliance on manual labor, improving crop monitoring, and promoting sustainable farming practices.

Methodology

A Fig. 1 provides an overview of the proposed system for MSD. It integrated three modules involving *Detection algorithm*, *mature strawberry detection* and a *deployment*. This integrated strategy addresses the challenges of accurate and real-time crop monitoring. The deployment was carried out in both simulation and real-time modes to validate the proposed algorithm.

Detection model

Effective crop monitoring is essential prior to harvesting ripe strawberries. To accurately detect ripe strawberries,

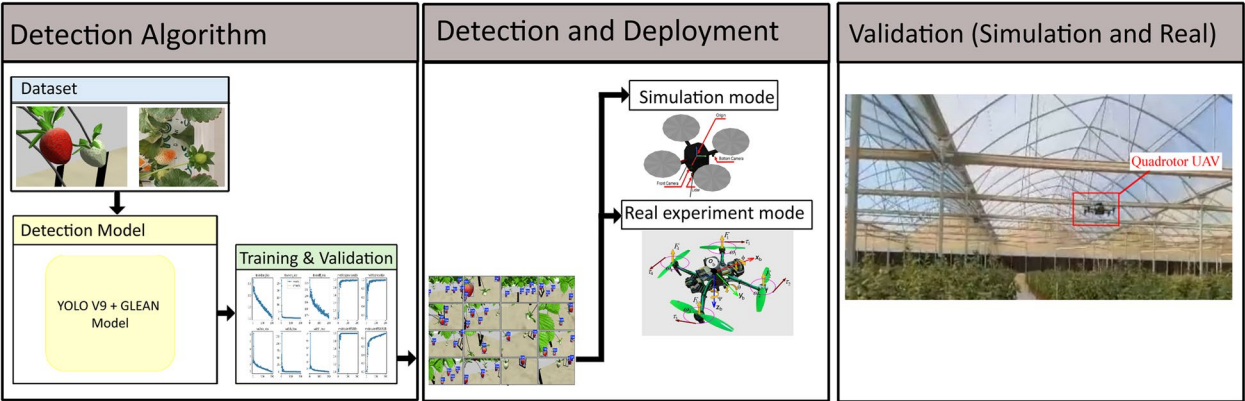


Fig. 1 The Strawberry detection architecture is deployed on a quadrotor system for practical, real-world applications. The quadrotor system includes a camera to facilitating automated crop monitoring of strawberry plants

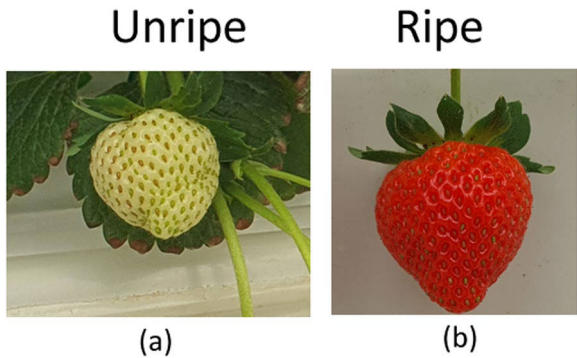


Fig. 2 Strawberry fruit. **a** an unripe fruit, and **b** a fully ripe fruit

it is crucial to first study the fruit’s classes (unripe and ripe) to collect corresponding data. This dataset serves as the foundation for training an accurate detection model. In Fig. 2 represents a strawberry classes (a) an unripe fruit, and (b) a fully ripe fruit.

Strawberries, cherished for their vibrant color, sweet flavor, and rich aroma, are not only a popular fruit worldwide but also a powerhouse of nutrients. Given their delicate structure, perishable nature, and visual cues associated with ripeness, strawberries are an ideal candidate for precision agriculture applications. Computer vision technologies can significantly aid in monitoring plant health, assessing fruit maturity, and automating harvest timing. By integrating AI-powered systems into strawberry farming, growers can optimize yield, reduce labor costs, and ensure consistent fruit quality, ultimately promoting sustainable and efficient agricultural practices.

Dataset

A quadrotor was utilized to capture crop data in video format during flying over the crop, and the detection model was later applied in offline mode to analyze the number of ripe strawberries. In this study, we utilized a dataset comprising 250 images from a CAD-based

Table 1 Amount of data to train the detection model: The dataset is created using real time strawberries images and synthetic CAD images

Classes	Greenhouse farm	Synthetic CAD
Ripe	800	250

synthetic dataset and 800 images for real strawberries in greenhouse. The variation in the dataset enhance the model performance validity. So, total 1050 images were used in dataset. The dataset consisted of one class: ripe strawberries (0). The domain randomization technique is applied on the images by variations in brightness, different backgrounds, and shaded areas. Table 1 provides comprehensive details about the images contained within the dataset. To enhance training data diversity, various augmentation methods were implemented, including rotation, flipping, and blurring transformations. These techniques contribute to strengthening the model’s generalization capabilities across diverse operational conditions. The dataset was divided using a standard 70:20:10 split ratio, allocating 735 images for training, 210 images for validation, and 105 images for testing. This distribution ensures sufficient training data while maintaining adequate validation and test sets for reliable performance evaluation.

Figure 3 presents a comparative view of both the actual strawberry farm image dataset environment and the synthetic Gazebo environment generated through CAD methodology using Blender software. Within this dataset, mature strawberries exhibit distinctive features including a rich red coloration, smooth texture, and substantial, well-developed fruit size. These defining characteristics serve as essential indicators for determining strawberry maturity and provide valuable resources for researchers and developers working on automated strawberry ripeness detection systems, ultimately contributing to improved operational efficiency and product quality in the strawberry cultivation industry.

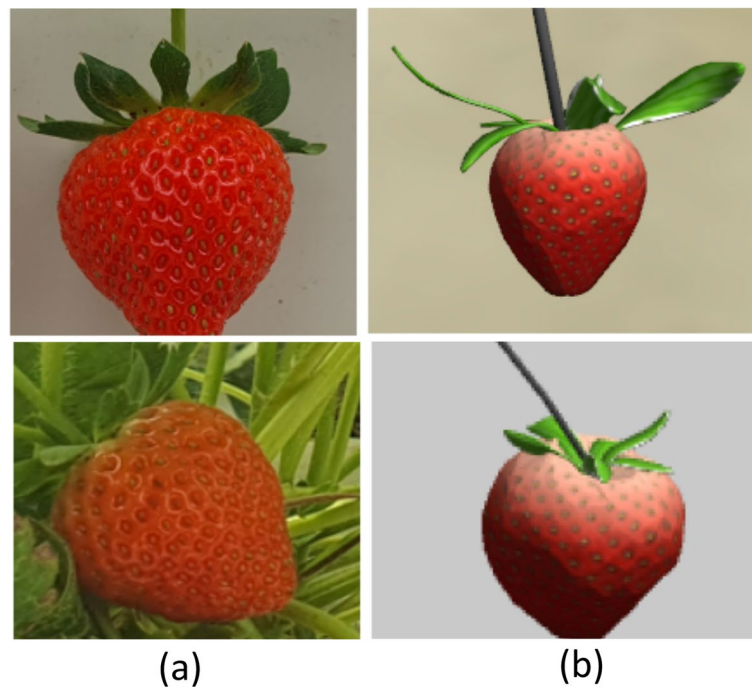


Fig. 3 Image dataset collection environments (a) Real ripe strawberries dataset, and (b) CAD synthetic ripe strawberries dataset created using Blender software

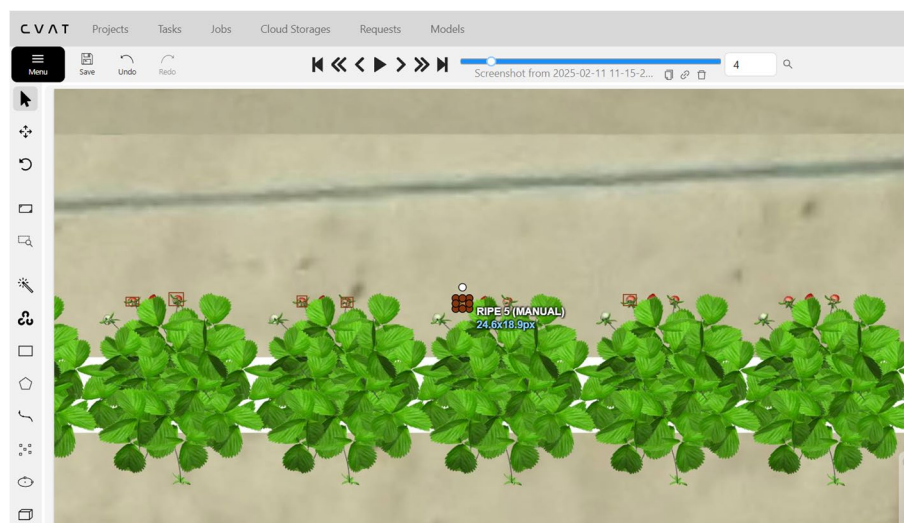


Fig. 4 Image dataset collection environments (a) Real ripe strawberries dataset, and (b) CAD synthetic ripe strawberries dataset created using Blender software

The Computer Vision Annotation Toolbox (CVAT) was used for markup. It allows you to make annotations in various formats. Figure 4 shows an example of data markup in CVAT [36].

Detection model architecture

Conventional image analysis techniques rely on scanning datasets through numerous sliding windows to extract pertinent features, followed by classifier implementation for object identification. To enhance the effectiveness of

this methodology, researchers developed a CCL-SVM framework [71] utilizing Support Vector Machine algorithms specifically designed for identifying diseases in tomato foliage, particularly under complex and demanding environmental conditions. A fruit detection and quality assessment system based on YOLOv5 was introduced by Goyal (2022) [23], operating through a two-phase process. The primary detection stage attained a mean Average Precision of 92.80% for identifying fruits. During the quality analysis phase, the model recorded mAP scores of

99.60% and 93.10% for apples and bananas respectively. This method proved highly effective in automated fruit sorting applications, consistently producing accurate classification results. An innovative strawberry ripeness assessment technique was presented by Yang (2023) [70], which merged YOLOv8 with the LW-Swin Transformer for improved detection capabilities. The team modified the feature fusion network by integrating a residual structure containing learnable parameters and scaled normalization elements into the standard Swin Transformer design. Validation results confirmed that their proposed LS-YOLOv8s model substantially exceeded competing methods, demonstrating mAP@0.5 improvements of 1.6% over YOLOv5s, 33.5% over CenterNet, and 3.4% over SSD during experimental evaluation.

In our research we are using YOLOv9 base model instead of the more recent YOLOv10, v11, or v12 due to following reasons:

- YOLOv9 introduces a strong balance between accuracy, inference speed, and model size, making it suitable for real-time embedded or edge deployments. While YOLOv10–v12 may have improved accuracy, they often come with increased computational complexity or larger model sizes that are not ideal for all environments (e.g., low-power robots, drones, IoT devices) [66]

- In our tasks such as (e.g., greenhouse strawberry detection using both synthetic CAD and real images), YOLOv9 handles domain shifts effectively due to its strong generalization in mixed-data scenarios.

The research proposes a new architectural concept known as Generalized ELAN (GELAN), representing an advanced iteration of the original ELAN framework [25]. By integrating CSPNet and ELAN concepts, GELAN enables strategic gradient pathway management. The architecture's development focuses on achieving minimal computational overhead, accelerated inference performance, and enhanced accuracy levels. The overall architecture of GELAN is depicted in Fig. 5. To assess GELAN's effectiveness, we performed experiments using a mixed dataset containing both synthetic CAD-generated images and real strawberry photographs. These experiments took place in a controlled gazebo greenhouse setting. The experimental findings demonstrated that our proposed YOLOv9 model outperformed all other compared models across every evaluation metric tested.

In this study, we trained and validated a detection model using an enhanced YOLOv9 architecture integrated with the GELAN framework for strawberry crop monitoring applications. As shown in Fig. 6, the

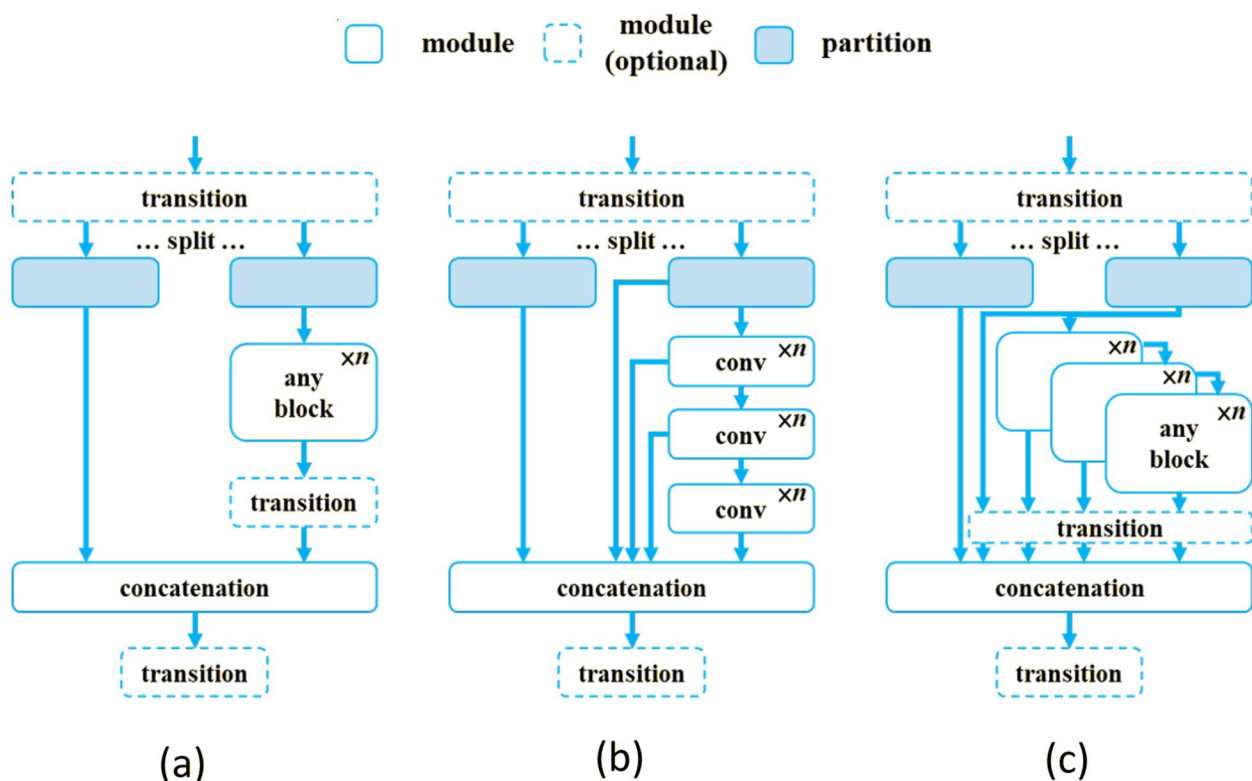


Fig. 5 The architecture of GELAN: **a** CSPNet, **b** ELAN, and **c** GELAN [66]

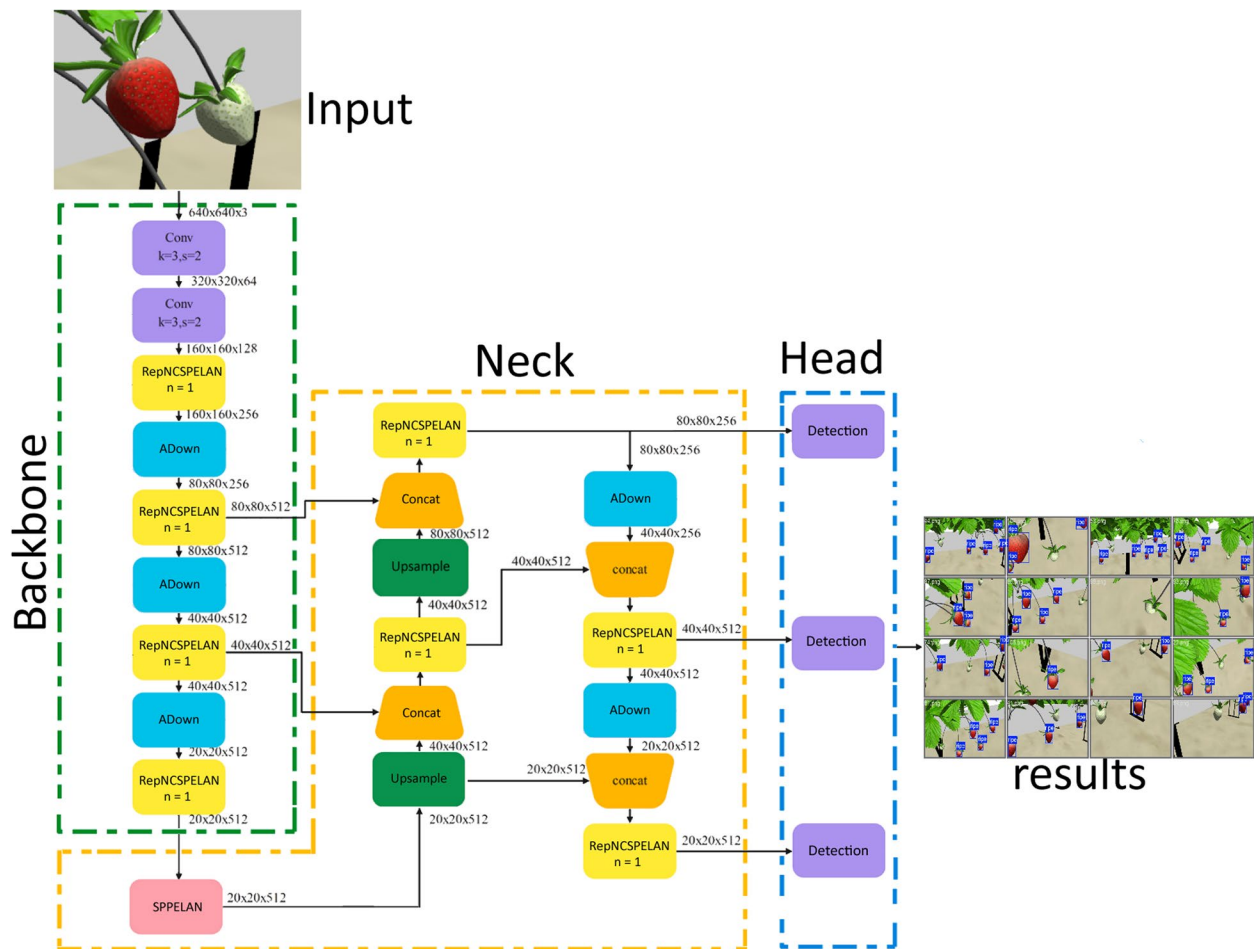


Fig. 6 YOLOv9-GELAN architecture for strawberry detection, featuring a backbone network (green) for feature extraction through RepNCSPPELAN blocks, a neck network (yellow) with Feature Pyramid Network for multi-scale feature fusion, and detection heads (blue) that output precise bounding boxes for ripe strawberry detection

YOLOv9-GELAN architecture is structured into three primary components: the backbone, neck, and head. The model accepts input images of size $640 \times 640 \times 3$, representing RGB color channels. Initially, two convolutional blocks with a kernel size of 3 and a stride of 2 are applied to reduce the spatial resolution. The number of channels in these convolution blocks varies based on the configuration of the YOLOv9 architecture.

The output from the initial convolution layers is passed to the RepNCSPPELAN block, which defines the number of replications and bottleneck blocks used. Additionally, an ADown block is employed for downsampling, further reducing the feature map resolution. At the end of the backbone, the RepNCSPPELAN block connects to the SPPELAN (Spatial Pyramid Pulling Efficient Layer Aggregation Networks) block. In the neck section, the SPPELAN output is upsampled to align with the feature map resolution of the RepNCSPPELAN block using an Upsample block. A Concat block then merges the features from both the Upsample and RepNCSPPELAN

blocks by summing the channel values while preserving spatial dimensions. This alignment ensures that the feature resolutions from both the backbone and neck components are consistent. Subsequently, the feature map is forwarded to the head of the architecture, which consists of three detect blocks.

Each block is designed to handle objects of different sizes: small, medium, and large. These blocks operate on feature maps with a resolution of $20 \times 20 \times 512$. The training is done for 300 epochs with batch size 16 and image size 640.

Training process

The network training for this research was conducted using a hardware setup that included an Intel(R) Core(TM) i9-14900KF processor operating at 3.20 GHz, 128 GB of system memory, a 2 TB hard disk drive, and an NVIDIA GeForce RTX 3090 graphics processing unit. Ubuntu served as the operating system platform, with PyTorch functioning as the deep learning framework.

The software environment comprised Python version 3.9.18, torch-2.1.2+cu118, and OpenCV 3.4.5. Training was performed on YOLOv9-GELAN, YOLOv9, and YOLOv8 models using input images scaled to 640 x 640 pixels. The training configuration employed a starting learning rate of 0.001, momentum parameter of 0.90, weight decay coefficient of 0.0005, and a batch size of 16 samples. All models underwent training for 300 epochs, with multiple iterations through the complete dataset to achieve optimal model performance.

The trained model was evaluated on validation dataset using various evaluation metrics to assess their performance. These metrics included precision (P), recall (R), F1 score, average precision (AP), and mean average precision (mAP). The Mean Average Precision (mAP) statistic is often used to evaluate object-detecting algorithms. This measure is computed by comparing predicted bounding boxes to their corresponding ground truth bounding boxes. The Intersection Over Union (IoU) is an important component of this comparison since it evaluates the overlap between the anticipated bounding box and the ground truth.

$$IoU = \frac{\text{area of overlap}}{\text{area of union}} \quad (1)$$

Precision is the proportion of true positives out of all the positive predictions made by a model. It measures how accurate the model's positive predictions are.

$$Precision = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

Recall is the proportion of true positive samples that the model correctly identified as positive out of all the true positive samples. In other words, recall measures how well the model is able to find or detect the positive samples. A higher recall value indicates the model is able to correctly identify a larger proportion of the actual positive cases.

$$Recall = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

The F1 score combines these two factors into a single value that indicates how well the model can accurately predict outcomes. An improved ability to balance accurately detecting instances that are positive with reducing false positives is shown by a higher F1 score for the model. To summaries, the F1 score evaluates a classification model's overall efficacy by considering its precision (accuracy in producing positive predictions) and recall (ability to identify positive occurrences). It is a useful

metric for evaluating how well a model can anticipate outcomes.

$$F1_{score} = 2 \times \frac{Recall \times Precision}{Recall + Precision} \quad (4)$$

Figure 7 illustrated the confusion matrix for proposed strawberry detection model YOLOv9-GLEAN and other baseline models YOLOv8 and YOLOv9, comparing their performance in classifying ripe strawberries versus background elements. Each matrix displays the relationship between true labels (horizontal axis) and predicted labels (vertical axis) for binary classification tasks. Figure 7a shows the YOLOv8 model's performance with 735 total predictions. The model correctly identified 711 ripe strawberries (true positives) and misclassified 22 ripe strawberries as background (false negatives). Additionally, 3 background elements were incorrectly classified as ripe strawberries (false positives). This demonstrates high sensitivity for ripe strawberry detection with minimal false positive errors. Figure 7b presents YOLOv9 performance with 735 total predictions. The model achieved 719 correct ripe strawberry classifications (true positives) with only 15 false negatives, representing better recall compared to model Fig. 7a. The false positive rate remained extremely low with only 1 background misclassifications, indicating enhanced precision. Whereas, Fig. 7c exhibits the best overall performance (proposed YOLOv9-GLEAN) with 735 predictions. This model correctly classified 724 ripe strawberries (highest true positive rate) while reducing false negatives to just 10 cases. The false positive rate remained minimal with only 1 background misclassifications, demonstrating optimal balance between precision and recall.

The training progress analysis reveals exceptional performance of the YOLOv9-GLEAN based strawberry detection model across 200 epochs of training. Figure 8 illustrated the training process performance parameters. The localization accuracy improved significantly, with the training box loss decreasing from an initial value of 2.6 to a final converged value of 1.0, while the validation box loss demonstrated similar improvement from 6.0 to approximately 1.0, indicating robust generalization capabilities. Classification performance showed remarkable convergence, with training classification loss rapidly declining from 28 to near-zero within the first 50 epochs, and validation classification loss exhibiting even more dramatic improvement from 140 to essentially zero. The Distribution Focal Loss (DFL) metrics further confirmed model reliability, with training DFL loss reducing from 2.2 to 1.7 and validation DFL loss decreasing from 7.0 to 1.8, demonstrating improved confidence calibration throughout the training process.

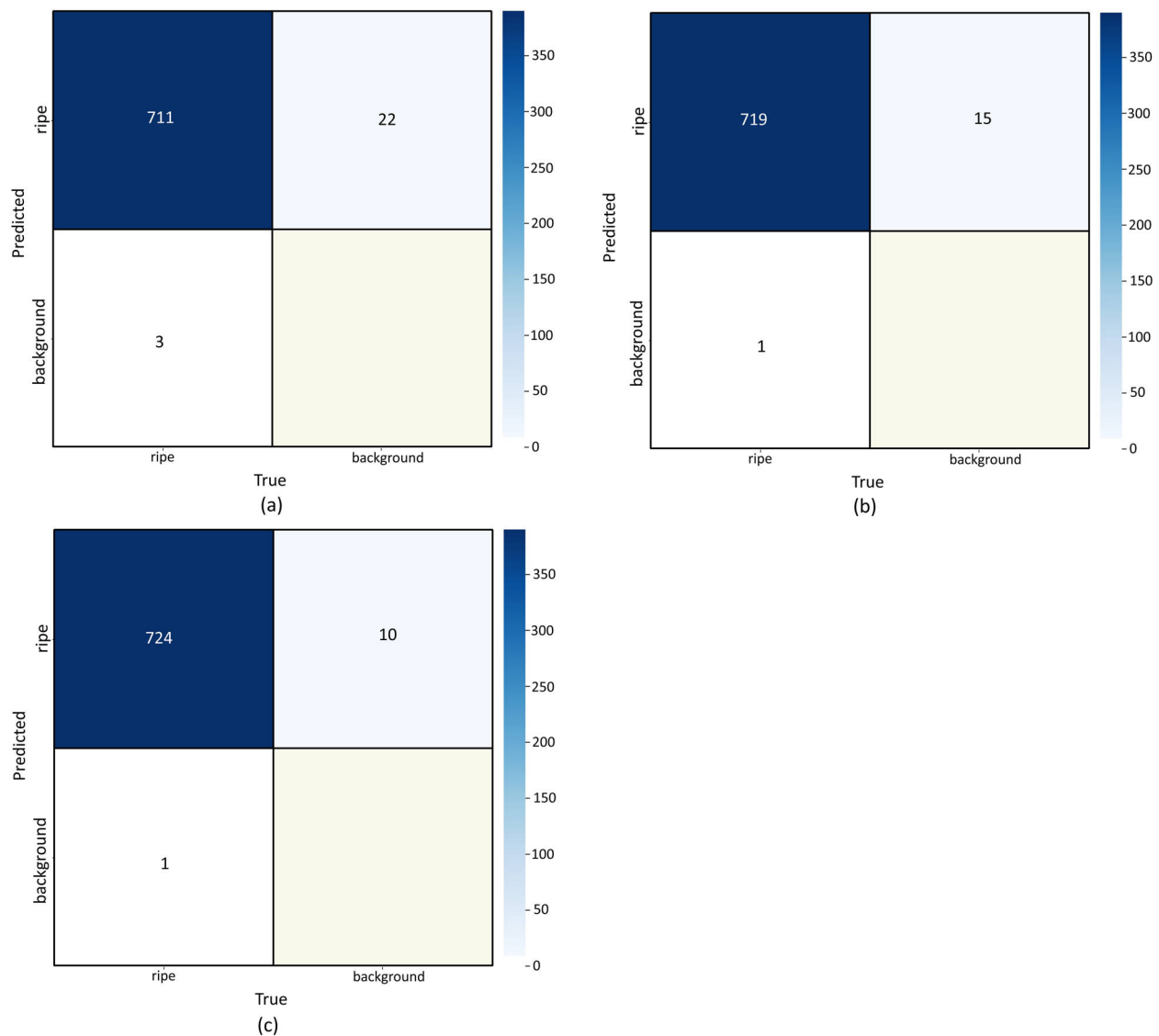


Fig. 7 Confusion matrix **a** YOLOv8, **b** YOLOv9, and **c** YOLOv9-GLEAN

The performance metrics underscore the model's exceptional detection capabilities, with precision rapidly ascending from 0.2 to nearly 1.0 within the initial 25 epochs and maintaining this high level throughout training. Recall performance followed a similar trajectory, achieving near-perfect scores of 1.0 early in training and sustaining this performance consistently. The mean Average Precision at IoU threshold 0.5 (mAP50) reached outstanding values of approximately 1.0, while the more stringent mAP50-95 metric, which evaluates performance across multiple IoU thresholds from 0.5 to 0.95, achieved impressive scores of 0.95. These metrics demonstrate that the model achieved rapid initial learning within the first 25-50 epochs, followed by stable convergence without signs of overfitting, making it highly

suitable for real-world automated strawberry detection and monitoring applications in agricultural settings.

Figure 9 illustrated the performance evaluation of the trained model. As depicted in Fig. 9 the model demonstrates exceptional performance in detecting ripe strawberries in video frames captured by quadrotor, achieving near-perfect precision and recall. In Fig. 9a F1-confidence curve indicates that the best balance between precision and recall occurs at an F1-score of 0.99 at a confidence threshold of 0.471.

The precision-confidence curve reveals that the model achieves a precision of 1.00 at a confidence threshold of 0.943 as per Fig. 9b, suggesting that when the confidence threshold is high, the model makes highly precise predictions with minimal false positives. In Fig. 9c the recall-confidence curve shows that the model attains 1.00 recall

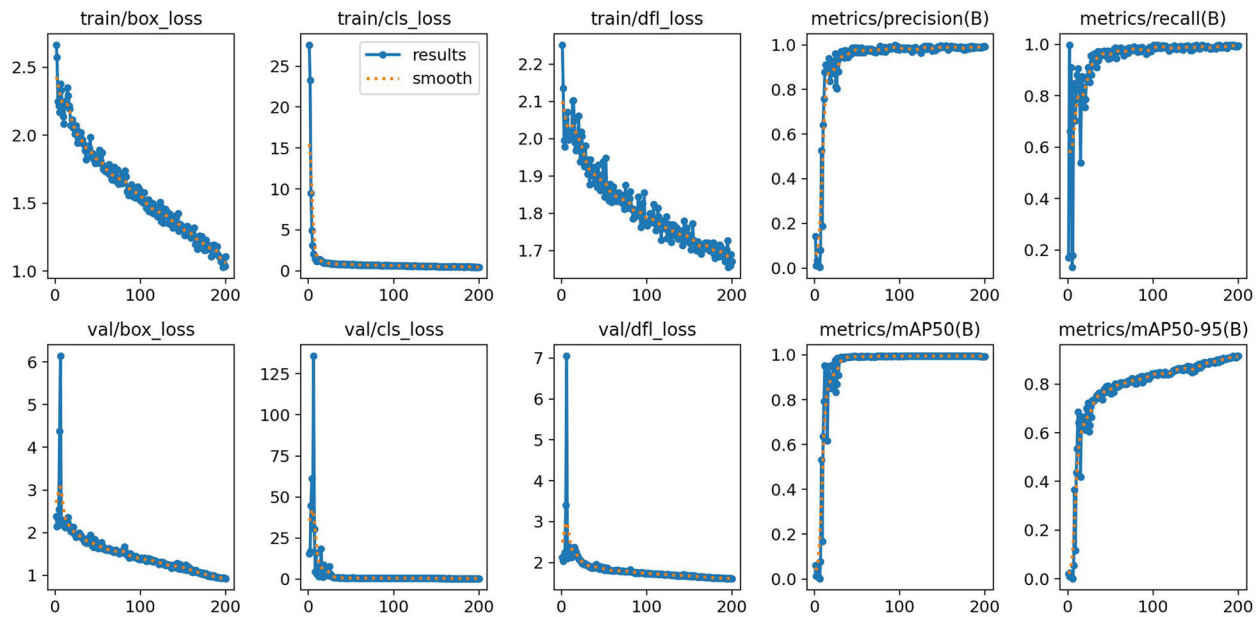


Fig. 8 Training process performance parameters

at a confidence threshold of 0.000, meaning it detects all ripe strawberries at low thresholds but may also include false positives. The precision-recall curve (Fig. 9d) further confirms the model's robustness, achieving a mean average precision (mAP@0.5) of 0.994, highlighting its high effectiveness in crop monitoring applications.

YOLOv9 extends the capabilities of YOLOv8 through the integration of advanced architectural components, including CSPNet (Cross-Stage Partial Networks) and ELAN (Efficient Layer Aggregation Networks). These components enhance the network's capacity to achieve an optimal balance between computational efficiency and detection accuracy across multiple object scales. CSPNet facilitates improved gradient propagation during the backpropagation process, resulting in more efficient learning while maintaining performance integrity. Meanwhile, ELAN enhances feature aggregation across different network layers, thereby strengthening overall feature extraction capabilities from input imagery. The key innovation in YOLOv9-GELAN lies in the incorporation of GELAN, which empowers the model to produce high-resolution image features that significantly enhance detection precision. This advancement proves particularly valuable for strawberry detection applications, where subtle visual characteristics such as color variations, surface texture, and fruit dimensions are critical for accurate ripeness classification. This capability is essential for strawberry detection in complex scenarios, including cluttered environments or reduced-quality images commonly encountered in greenhouse settings.

The precise details of the metrics and their respective scores can be found in Table 2. These results suggest that

the YOLOv9-GELAN model is more effective and robust in accurately detect tiny objects.

According to the findings in Table 2, the YOLOv9-GELAN model demonstrated superior performance compared to the other available models. YOLOv9-GELAN achieves a precision of 0.991 and recall of 0.990, outperforming previous and latest versions and also have more FPS (offline processing speed). The ability to better differentiate strawberries from the background is enhanced by GELAN's super-resolution capability, reducing false positives and negatives.

Quadrotor modeling

Although this study primarily focused on developing the MSD (Mature Strawberry Detection) algorithm for robust detection of ripe strawberries, our main research objective is to integrate this algorithm into a quadrotor platform to be used for greenhouse crop monitoring. Two main components of this robotic system include a system modeling, and hybrid trajectory tracking controller.

System modeling

Defining the coordinate frames is the first step in modeling quadrotor dynamics. Figure 10 shows the frames define in the quadrotor modeling. The quadrotor will operates in two frames (i) inertial frame (x_i, y_i, z_i) and body frame (x_b, y_b, z_b). The inertial frame is defined relative to the ground, with the positive z-axis oriented upward. In contrast, the body frame is defined with respect to the quadrotor's center of gravity, and its axes are fixed to the body of the quadrotor. The governing equation is derived by using projection matrices to

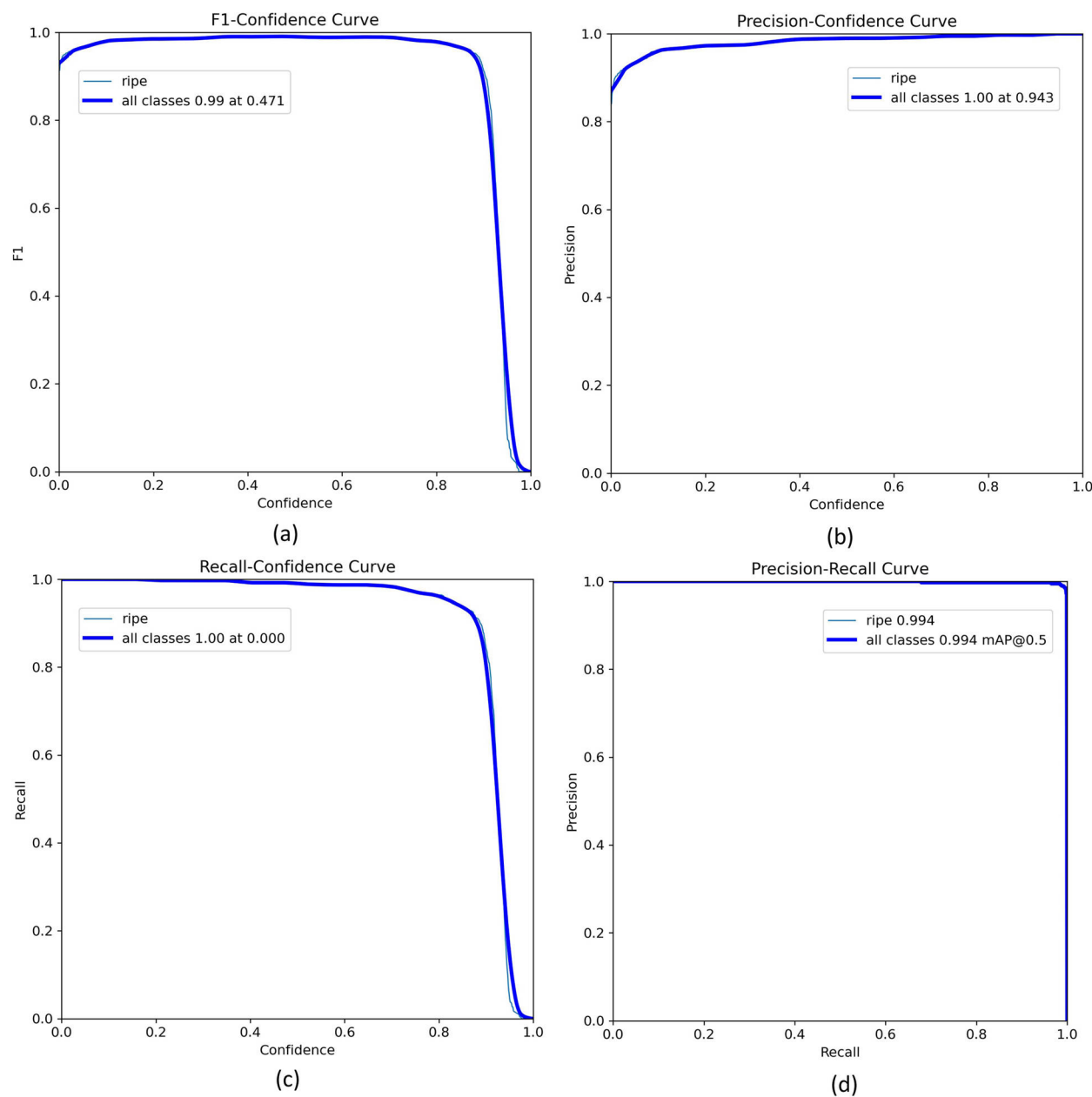


Fig. 9 Performance evaluation curves for YOLOv9-GELAN strawberry detection: **a** F1-confidence achieving 0.99 at threshold 0.471, **b** precision-confidence reaching 1.00 at 0.943, **c** recall-confidence maintaining 1.00 until 0.8 threshold, and **d** precision-recall curve with exceptional mAP@0.5 of 0.994, demonstrating superior detection performance

Table 2 Performance metrics comparison

Model	Params(M)	FLOPs	Precision	Recall	FPS	Size (mb)	Inference (ms)
Yolov12	24	21.5	0.987	0.98	27	45	2.0
Yolov11	14	21.6	0.981	0.99	26	27	1.9
Yolov10-s	11.3	26.0	0.99	0.992	25	21	1.5
Yolov8-s	11.2	25.6	0.98	0.979	28	22.5	0.9
Yolov9-s + GLEAN (proposed)	11.2	25.0	0.991	0.990	30	23	0.75

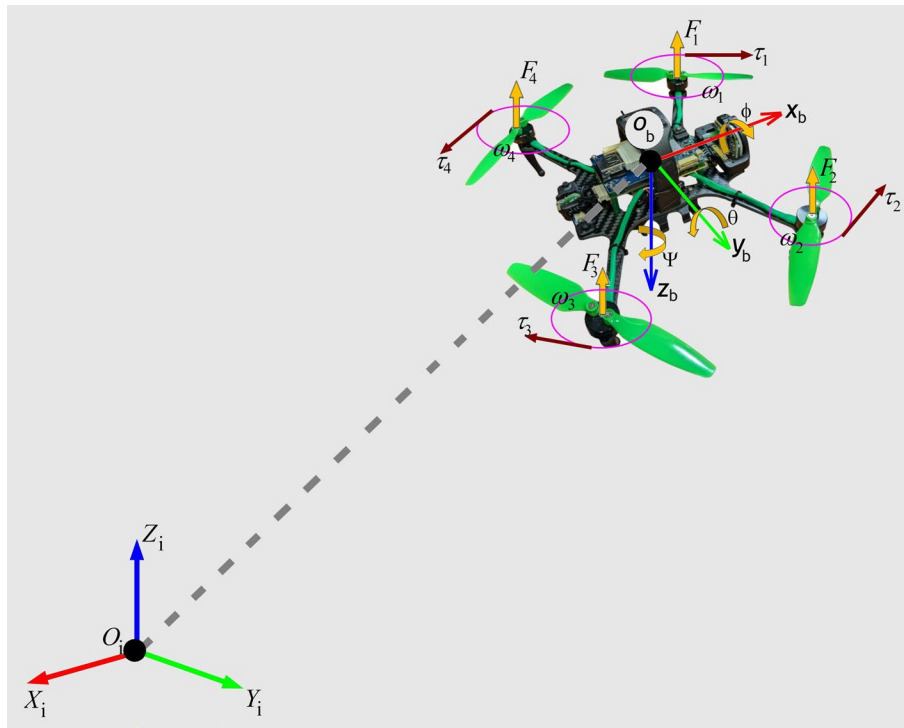


Fig. 10 Quadrotor diagram and its reference frames

transform quantities between the two frames [27]. Coordinate frame transformations are used to convert quantities between two reference frames. The quaternion method is employed to perform these transformations efficiently and accurately [34]. The coordinate transformation between two distinct reference frames is calculated using quaternion rotation as [22].

$$A = q \otimes A' \otimes q^{-1} \quad (5)$$

Here, A represents the rotated vector, and A' is the vector to be rotated. Equation 1 indicates that multiplying two or more rotation quaternions yields another quaternion. In pure quaternions, vectors can be transformed between frames by first converting them into quaternions with a zero scalar component. Thus, the quaternion-based rotation matrix that maps the inertial frame to the body frame can be derived by reformulating Eq. (5) as follows:

$$\begin{bmatrix} 0 \\ r_b \end{bmatrix} = q \otimes \begin{bmatrix} 0 \\ r_i \end{bmatrix} \otimes q^* \quad (6)$$

whereas $r_b = [x_b, y_b, z_b]'$ and $r_i = [x_i, y_i, z_i]'$ are position quantities with respect to body frame and inertial frame respectively. The 3D rotation matrix $R_B^I(q)$ for position projection from body frame to inertial frame is obtained using Eq. (6) as follows [54]:

$$\begin{bmatrix} x_i \\ y_i \\ z_i \end{bmatrix} = \begin{bmatrix} 2(q_o^2 + q_1^2) - 1 & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & 2(q_o^2 + q_2^2) - 1 & 2(q_2q_3 + q_0q_1) \\ 2(q_0q_2 + q_1q_3) & 2(q_2q_3 - q_0q_1) & 2(q_o^2 + q_3^2) - 1 \end{bmatrix} \begin{bmatrix} x_b \\ y_b \\ z_b \end{bmatrix} \quad (7)$$

The transformation matrix that projects the body frame rates $\omega = [p \ q \ r]'$ into the inertial frame rates $\dot{q}$ can be derived as follows [9]:

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} q \otimes \omega = \frac{1}{2} \begin{bmatrix} q_0 & -q_1 & -q_2 & -q_3 \\ q_1 & q_0 & -q_3 & q_2 \\ q_2 & q_3 & q_0 & -q_1 \\ q_3 & -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} 0 \\ p \\ q \\ r \end{bmatrix} \quad (8)$$

The following assumptions were made to model the quadrotor flight dynamics within the greenhouse environment.

- Quadrotor is rigid body with fixed mass and uniform symmetry.
- No external disturbance (wind) in the controlled greenhouse environment.
- Constant gravitational force.
- Motor resistance m_r is negligible.
- Pre-defined way points for quadrotor trajectory.

By defining the angular velocities for the four rotors of quadrotor as $\omega_1, \omega_2, \omega_3, \omega_4$, the total lift in the Z direction

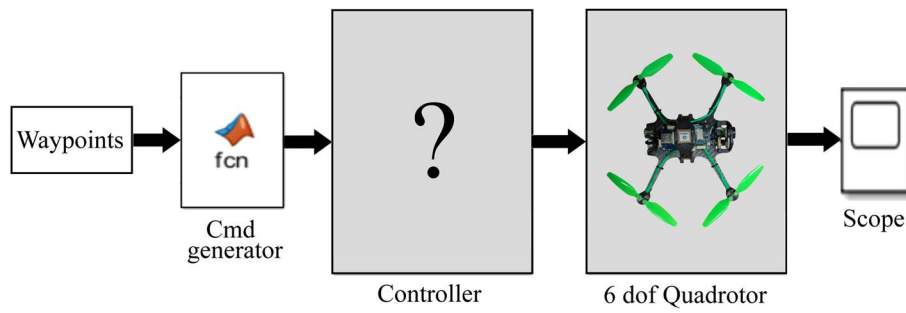


Fig. 11 Simulink model of 6 dof quadrotor for predefined waypoints

of body frame as T_L , and the torques around body frame axis as τ_x, τ_y, τ_z , and then it can be obtained as:

$$T_L = L_c (\omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2) \quad (9)$$

$$\tau_x = l L_c (\omega_2^2 - \omega_4^2) \quad (10)$$

$$\tau_y = l L_c (-\omega_1^2 + \omega_3^2) \quad (11)$$

$$\tau_z = b (-\omega_1^2 + \omega_2^2 - \omega_3^2 + \omega_4^2) \quad (12)$$

Here, L_c represents the lift coefficient, b denotes the anti-torque coefficient, and l corresponds to the length of the quadrotor arm. The model of the quadrotor is calculated as:

$$\ddot{x} = \frac{1}{m} [T_L (\cos\phi \sin\theta \cos\psi + \sin\phi \sin\psi) - d_c \dot{x} + F_{dx}] \quad (13)$$

$$\ddot{y} = \frac{1}{m} [T_L (\cos\phi \sin\theta \sin\psi - \sin\phi \cos\psi) - d_c \dot{y} + F_{dy}] \quad (14)$$

$$\ddot{z} = \frac{1}{m} [T_L \cos\phi \cos\theta - d_c \dot{z} - mg + F_{dz}] \quad (15)$$

$$\dot{\phi} = p + q \sin\phi \tan\theta + r \cos\phi \tan\theta \quad (16)$$

$$\dot{\theta} = q \cos\phi - r \sin\phi \quad (17)$$

$$\dot{\psi} = \frac{1}{\cos\theta} (q \sin\phi + r \cos\phi) \quad (18)$$

$$\dot{p} = \frac{1}{I_{xx}} [(I_{zz} - I_{yy}) qr + \tau_x + \tau_{dx}] \quad (19)$$

$$\dot{q} = \frac{1}{I_{yy}} [(I_{xx} - I_{zz}) pr + \tau_y + \tau_{dy}] \quad (20)$$

$$\dot{r} = \frac{1}{I_{zz}} [(I_{yy} - I_{xx}) qp + \tau_z + \tau_{dz}] \quad (21)$$

Table 3 Specifications of the quadrotor for mature strawberry detection

Component	Specification
Take-off Weights	285g
Flight Time	40 min
Motors rating	3000 kw
Propellers dia	120 mm
Camera	12 MP color image sensor
GPS	UBlox M10
Autonomy	Integrated Qualcomm QRB5165 chipset onboard
Software compatibility	OpenCV, ROS2, PX4

where x, y, z denotes the position in inertial frame, m mass of quadrotor, g is the gravity, d_c is drag coefficient; ϕ, θ, ψ represent the roll, pitch and yaw angle, p, q, r denotes the angular rates around body frame, I_{xx}, I_{yy} , and I_{zz} are moments of inertia; F_{dx}, F_{dy}, F_{dz} are external disturbance forces, and $\tau_{dx}, \tau_{dy}, \tau_{dz}$ denotes the external disturbance torques. In our case external disturbance forces and torques are negligible due to control environment of the greenhouse. Using Eqs. (13) to (21), a MATLAB-Simulink model was developed, as shown in Fig. 11.

Table 3 mentioned the hardware specifications of the quadrotor deployed for mature strawberry detection application.

Trajectory tracking controller design

Research in quadrotor controller design is a rapidly growing field driven by the rising demand for autonomous drones in areas such as delivery, surveillance, agriculture, and disaster response. Developing robust and efficient controllers for quadrotors presents significant challenges, including managing nonlinear dynamics, under-actuation, axis coupling, and external disturbances like wind. Controllers like PID [37, 53], Adaptive PID [39], LQR [17, 40], and MPC [18, 29] are extensively utilized for controlling linear and nonlinear quadrotor systems. In our work, no external disturbances, such as wind, are present since the greenhouse provides a controlled environment. Therefore, we proposed robust controller using classical PID and optimal LQR for the quadrotor, as they are

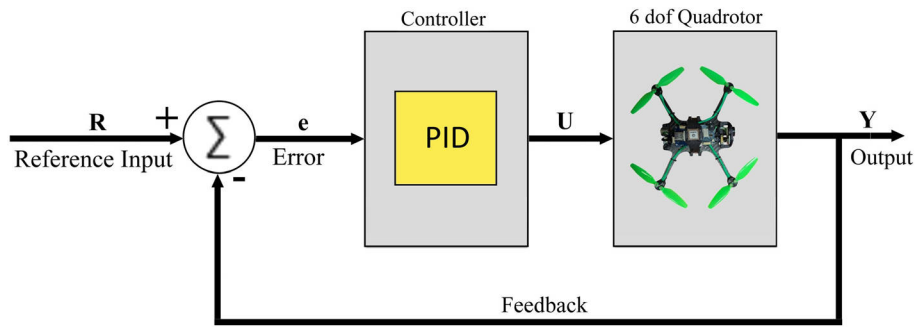


Fig. 12 Closed loop PID controller for Quadrotor system

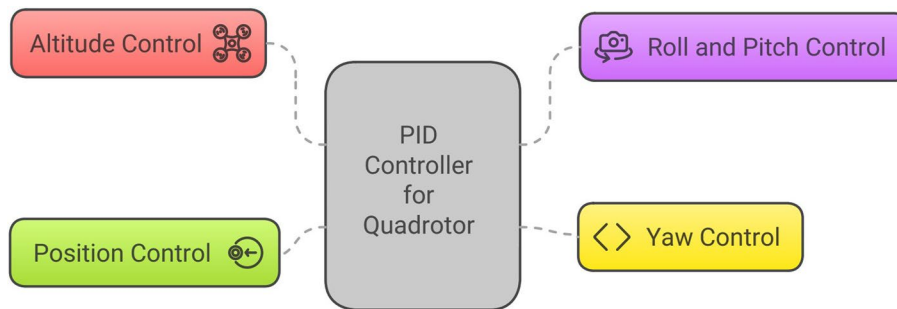


Fig. 13 Overall PID controller for quadrotor comprises of sub-control modules i.e. altitude control, roll and pitch control, position control, and yaw control

simple and easy to implement on the system. Combining PID and LQR controllers for quadrotor control offers both robustness and optimal performance. The proposed controller is designed to first achieve the target position in the Z direction (maintaining a constant height) and then follow the predefined way-points for scanning strawberry plants within the greenhouse.

- **PID controller** Designing a PID controller for a quadrotor requires managing its motion across multiple axes, including roll, pitch, yaw, and altitude. A quadrotor has six degrees of freedom (DOF): three translational (x, y, z) and three rotational (roll, pitch, yaw). The PID controller plays a crucial role in stabilizing and regulating these movements. The PID controller is a classic model in control theory, designed to minimize errors by adjusting the proportional, integral, and derivative coefficients in the quadrotor control equation. Figure 12 depicts the closed loop PID controller for quadrotor system. The proportional term adjusts the error by multiplying it with a proportionality constant, either increasing or decreasing the response. The derivative term predicts the controller's future behavior by analyzing the rate of change of error over time. The integral term accumulates all past error values, with the integral coefficient determining its influence to achieve the desired output in the shortest time. Therefore, tuning

coefficients proportional, integral, and derivative is essential to reach the control system's set-point effectively. Error (e) is defined as the difference between the reference input (R) value and the output of the system (Y) at any given moment. The output of the controller (U) is fed to the quadrotor system as input. The overall PID controller for the quadrotor comprises sub-control modules, including altitude control, roll and pitch control, position control, and yaw control, as illustrated in the Fig. 13. The general PID control equation is given below: whereas $U(t)$ is the control command to the system, K_p , K_i , K_d are proportional, integral and derivative gains respectively and $e(t)$ is the error value.

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt} \quad (22)$$

- **Altitude control:**
- The force in the upward Z-axis is given by:

$$F_z = m\ddot{z} + mg \quad (23)$$

where: F_z is the total thrust force along Z-axis, m is the mass of the quadrotor, $\ddot{z}$ is the upward acceleration, and g is the gravitational acceleration. The control objective is

to regulate the altitude (z) to track a desired altitude value (z_{desired}). The altitude error is defined as:

$$e_z = z_{\text{desired}} - z_{\text{actual}} \quad (24)$$

The control force (F_z) must be designed to ensure the error (e_z) converges to zero. The PID control law for altitude is expressed as:

$$F_z = m(g + \ddot{z}_{\text{desired}}) + K_p e_z + K_i \int e_z dt + K_d \frac{de_z}{dt} \quad (25)$$

where, $\ddot{z}_{\text{desired}}$ is desired vertical acceleration. The dynamics of the altitude error is:

$$\ddot{z} = \frac{F_z}{m} - g \quad (26)$$

Substitute the value of control law F_z from Eqs. (21) in (22):

$$\ddot{z} = \frac{1}{m} \left[m(g + \ddot{z}_{\text{desired}}) + K_p e_z + K_i \int e_z dt + K_d \frac{de_z}{dt} \right] - g \quad (27)$$

$$\ddot{z} = \ddot{z}_{\text{desired}} + \frac{1}{m} \left(K_p e_z + K_i \int e_z dt + K_d \frac{de_z}{dt} \right) \quad (28)$$

The error dynamics is:

$$\ddot{e}_z + \frac{K_d}{m} \dot{e}_z + \frac{K_p}{m} e_z + \frac{K_i}{m} \int e_z dt = 0 \quad (29)$$

- Roll (ϕ) and Pitch (θ) control:

The roll (ϕ) and pitch (θ) controllers manage the rotational dynamics of the quadrotor along the x and y axes, with motor torques playing a crucial role in regulating these dynamics. Euler's equation for rotational motion for roll (ϕ) and pitch (θ) is as follow:

$$I_{xx} \ddot{\phi} = \tau_\phi \quad (30)$$

$$I_{xx} \ddot{\theta} = \tau_\theta \quad (31)$$

where, I_{xx} and I_{yy} are the moments of inertia about the x and y axes, τ_ϕ and τ_θ are the torques generated by the motors, and $\ddot{\theta}$ and $\ddot{\phi}$ are the angular accelerations. The roll and pitch angle error is defined as:

$$e_\phi = \phi_{\text{desired}} - \phi_{\text{actual}} \quad (32)$$

$$e_\theta = \theta_{\text{desired}} - \theta_{\text{actual}} \quad (33)$$

The control torque for roll is given by:

$$\tau_\phi = K_{p\phi} e_\phi + K_{i\phi} \int e_\phi dt + K_{d\phi} \frac{de_\phi}{dt} \quad (34)$$

Substitute the τ_ϕ in Eq. (30).

$$I_{xx} \ddot{\phi} = K_{p\phi} e_\phi + K_{i\phi} \int e_\phi dt + K_{d\phi} \frac{de_\phi}{dt} \quad (35)$$

The roll angle error dynamics, represented by a second-order differential equation, governs the stabilization of the roll angle as follows:

$$\ddot{\phi} + \frac{K_{d\phi}}{I_{xx}} \dot{\phi} + \frac{K_{p\phi}}{I_{xx}} e_\phi + \frac{K_{i\phi}}{I_{xx}} \int e_\phi dt = 0 \quad (36)$$

Similarly, the control torque for pitch is:

$$\tau_\theta = K_{p\theta} e_\theta + K_{i\theta} \int e_\theta dt + K_{d\theta} \frac{de_\theta}{dt} \quad (37)$$

Substitute the τ_θ in Eq. (33).

$$I_{yy} \ddot{\theta} = K_{p\theta} e_\theta + K_{i\theta} \int e_\theta dt + K_{d\theta} \frac{de_\theta}{dt} \quad (38)$$

The pitch angle error dynamics, represented by a second-order differential equation, governs the stabilization of the pitch angle as follows:

$$\ddot{\theta} + \frac{K_{d\theta}}{I_{yy}} \dot{\theta} + \frac{K_{p\theta}}{I_{yy}} e_\theta + \frac{K_{i\theta}}{I_{yy}} \int e_\theta dt = 0 \quad (39)$$

The Eqs. (30) and (33) represents the final torques for roll and pitch angles control. Similarly, the Yaw and positional control torques for quadrotor are as below respectively.

- Yaw control torque (τ_ψ) :

$$\begin{aligned} \tau_\psi = & K_{p\psi} (\psi_{\text{desired}} - \psi_{\text{actual}}) \\ & + K_{i\psi} \int (\psi_{\text{desired}} - \psi_{\text{actual}}) dt \\ & + K_{d\psi} \frac{d(\psi_{\text{desired}} - \psi_{\text{actual}})}{dt} \end{aligned} \quad (40)$$

- Overall Position control: Vertical Axis.

$$F_z = m(g + \ddot{z}_{\text{desired}}) + K_{pz} e_z + K_{iz} \int e_z dt + K_{dz} \frac{de_z}{dt} \quad (41)$$

Horizontal Axes.

$$\ddot{x}_{\text{desired}} = K_{px} e_x + K_{ix} \int e_x dt + K_{dx} \frac{de_x}{dt} \quad (42)$$

$$\ddot{y}_{desired} = K_{py}e_y + K_{iy} \int e_y dt + K_{dy} \frac{de_y}{dt} \quad (43)$$

where, $e_x = (X_{desired} - X_{actual})$, $e_y = (Y_{desired} - Y_{actual})$, $e_z = (Z_{desired} - Z_{actual})$. The overall Simulink based PID controller for quadrotor is shown in the Fig. 14.

- LQR controller LQR (Linear Quadratic Regulator) is an optimal linear control algorithm designed to operate within a dynamic system by minimizing a predefined cost function. It is typically employed in scenarios where the system's parameters are fully known, and external disturbances, such as wind, that could impact the system are negligible. In our research, the greenhouse environment parameters are well-defined, and there are no external disturbances, such as wind. Therefore, utilizing an LQR controller for quadrotor applications within the greenhouse environment is a suitable choice. Compared to a classical PID controller, the LQR controller is more computationally complex due to the need to calculate the full state feedback

matrix K . This involves solving the Riccati equation, which requires more computational effort than the straightforward tuning of proportional, integral, and derivative gains in a PID controller. The LQR controller employs an optimal control algorithm designed to minimize a cost function derived from the system's equations. This cost function incorporates both the state variables and the input parameters, weighted by the Q and R matrices, respectively, to achieve the desired balance between state regulation and control effort. The state vector x for quadrotor is derived from Eqs. (9) to (14), is given by $x = (\phi, \theta, \psi, \dot{\phi}, \dot{\theta}, \dot{\psi}, x, y, z, \dot{x}, \dot{y}, \dot{z})$ and the input vector u is $u = (T_1, T_2, T_3, T_4)^T$. The two primary quantities to optimize in the quadrotor model are target reaching accuracy and response speed. To achieve a faster controller response, the values in the Q matrix should be adjusted, while minimizing the target reach setpoint error requires tuning the values in the R matrix Islam et al. [28]. The LQR cost function which needs to be optimized is given by Purnawan et al. [43]:

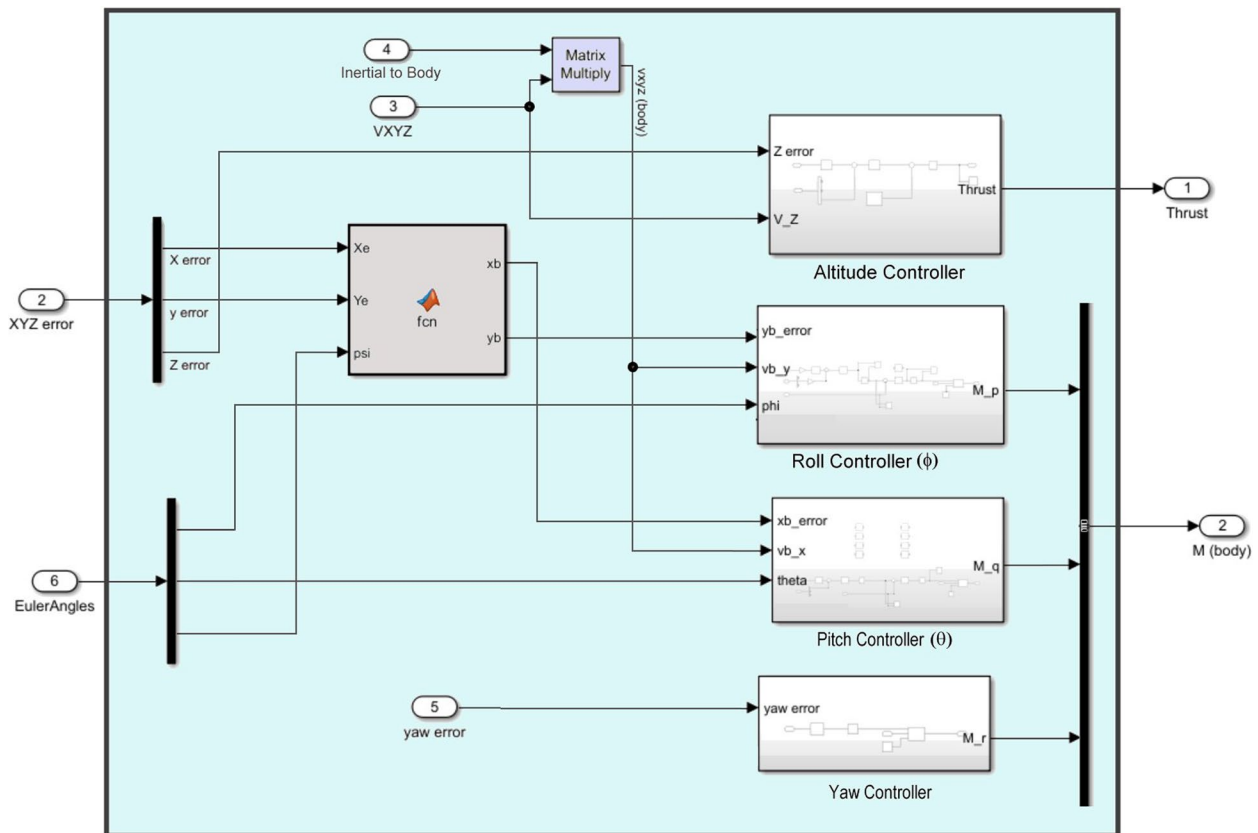


Fig. 14 The overall Simulink model for the quadrotor incorporates a PID controller, which regulates the system by adjusting the control inputs based on the error difference between the desired and actual states

$$J = \int (X^T Q X + U^T R U) dt \quad (44)$$

where, Q and R are positive definite symmetric matrices. By applying the optimality principle, the objective of the Linear Quadratic Regulator is to minimize the cost function provided in Eq. (40), which simplifies to a Riccati equation.

$$A^T M + M A - M B R^{-1} B^T M + Q = 0 \quad (45)$$

Here, M is a positive definite symmetric matrix. The optimal state feedback gain matrix is derived as follows.

$$K = R^{-1} B^T M \quad (46)$$

The final feedback control that drives the cost function towards zero is as below:

$$U = R - K X \quad (47)$$

Here, R represents the reference input, and X denotes the state vector of the system. Figure 15 illustrates the closed-loop control architecture used for designing the LQR controller. To design LQR, the quadrotor dynamics is linearized at a hovering position. The roll and pitch are not considered at hovering position. The optimized Q and R matrices for quadrotor system are defined as follows:

$$Q = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, R = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (48)$$

The optimal state feedback gain matrix K can be determined as follows.

$$K = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1.18 \\ 6.7 & 0 & 0 & 0 & 1 & 0 & 1.5 & 0 & 0 & 0 & 1.5 & 0 & 0 \\ 0 & 6.7 & 0 & -1 & 0 & 0 & 0 & 1.5 & 0 & -1.5 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1.14 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (49)$$

- Combining PID and LQR controllers for quadrotor control offers both robustness and optimal performance [6]. PID handles steady-state errors and disturbances effectively, making it ideal for outer-loop control like altitude or position. LQR, being model-based, provides fast and smooth responses for inner-loop tasks such as attitude stabilization. However, LQR can be sensitive to model inaccuracies. By blending both, the system benefits from PID's simplicity and disturbance rejection, along with LQR's optimal feedback control. This hybrid approach ensures improved stability, responsiveness, and robustness, making it well-suited for real-world quadrotor applications i.e. greenhouse environment. The total control law U_T :

$$U_T = U_{PID} + U_{LQR} \quad (50)$$

or explicitly:

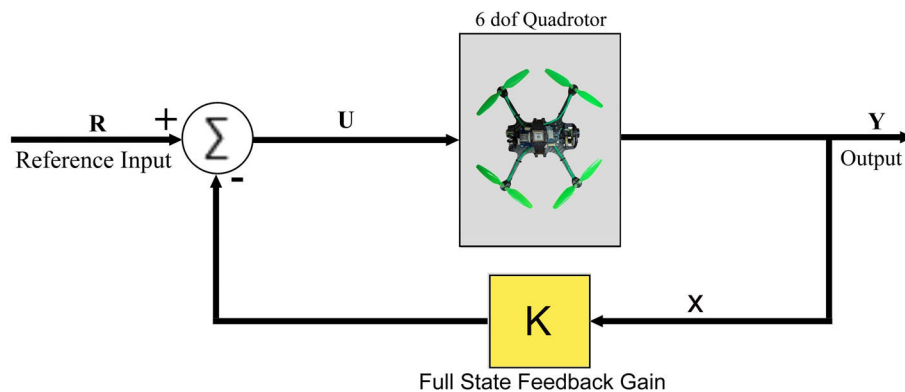


Fig. 15 The quadrotor system operates as a closed-loop system when integrated with an LQR controller. This setup ensures that the controller continuously adjusts the system's inputs based on feedback from the state variables, optimizing performance by minimizing the defined cost function

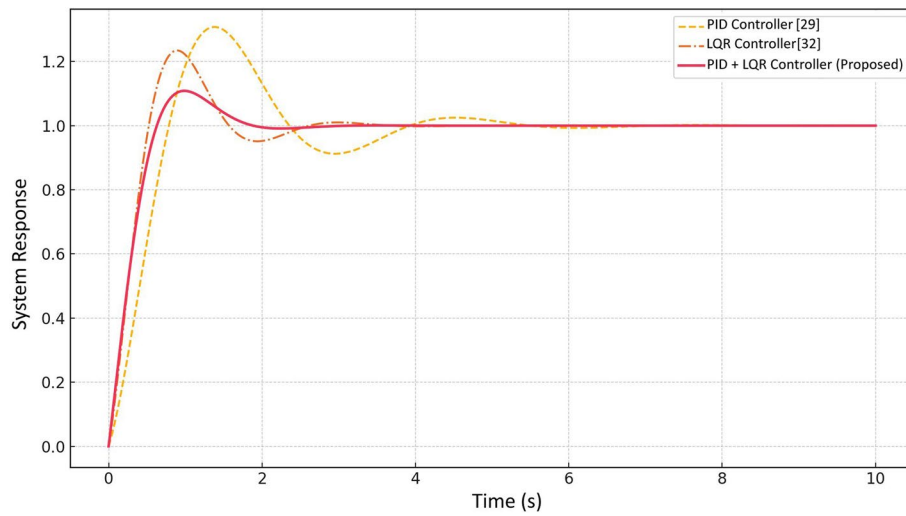


Fig. 16 The performance of proposed controllers is evaluated and compared with state of art conventional controllers using a step response analysis

Table 4 State of art step response comparison of the proposed controller

Controller	Overshoot(%)	Rise time(s)	Settling time(s)
PID [29]	30.77	0.79	0.77
LQR [31]	23.44	0.53	0.51
PID+LQR (Proposed)	10.85	0.63	0.61

$$U_T = K_p e + K_i \int e dt + K_d \frac{de}{dt} - K_{LQR} X \quad (51)$$

Therefore combined control law for yaw controller (τ_ψ) is depicted in Eq. (52)

$$\tau_\psi = K_{p\psi} e_\psi + K_{i\psi} \int e_\psi dt + K_{d\psi} \frac{de_\psi}{dt} - K_{LQR,\psi} X_\psi \quad (52)$$

Similarly, Eqs. (53) and (54) states the control law for Roll (ϕ) and Pitch (θ) control of quadrotor.

$$\tau_\phi = K_{p\phi} e_\phi + K_{i\phi} \int e_\phi dt + K_{d\phi} \frac{de_\phi}{dt} - K_{LQR,\phi} X_\phi \quad (53)$$

$$\tau_\theta = K_{p\theta} e_\theta + K_{i\theta} \int e_\theta dt + K_{d\theta} \frac{de_\theta}{dt} - K_{LQR,\theta} X_\theta \quad (54)$$

where, $X_\phi = \begin{bmatrix} \phi \\ \dot{\phi} \end{bmatrix}$, includes the roll angle and roll rate.

and $X_\theta = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$, includes the pitch angle and pitch rate.

The system performance of PID, LQR and PID + LQR (proposed) controllers are evaluated and compared using a step response analysis and mentioned in Fig. 16. Table 4 shows the controller performance comparison. The PID controller exhibited the highest overshoot (30.77%) and the slowest rise time (0.79 s) and settling time (0.77 s). Although it achieved a near-zero steady-state error

(0.0001), its response was less stable with notable oscillations. The LQR controller performed better with lower overshoot (23.44 %), faster rise time (0.53 s), and quicker settling time (0.51 s), achieving zero steady-state error. However, some oscillations were still present. The proposed PID + LQR controller outperformed both, with the lowest overshoot (10.85%), good rise time (0.63 s), and stable settling time (0.61 s). It also maintained zero steady-state error, indicating a fast, smooth, and accurate response. The proposed controller provides the most balanced and stable control performance, effectively reducing overshoot while maintaining fast response and accuracy—making it the most suitable choice for robust quadrotor control.

While the proposed hybrid PID-LQR controller demonstrates superior trajectory tracking performance, several limitations constrain its practical implementation. Dual-controller architecture increases computational complexity. The system exhibits significant sensitivity to model uncertainties, including center of gravity shifts from mounted equipment and motor parameter variations due to battery discharge or temperature changes. Additionally, in future the tuning of controllers against high-frequency disturbances common in greenhouse environments, such as sudden ventilation-induced air currents exceeding 2 m/s will be considered.

Experiments

This section presents the implementation of the proposed MSD object detection algorithm on quadrotor platforms through both simulated and real-world testing environments. Initially, a simulation framework was established using a digital twin representation of a strawberry greenhouse facility. Within this virtual environment, the quadrotor navigated above strawberry crops to identify mature fruit and conduct agricultural

monitoring. Subsequently, field trials were conducted in actual strawberry greenhouse operations using physical quadrotor systems to verify the effectiveness and practical viability of the developed approach.

Simulation framework

The integration of quadrotor UAVs (Unmanned Aerial Vehicles) in greenhouse environments has revolutionized plant monitoring by providing autonomous, high-precision data collection with minimal human intervention. Traditional ground-based monitoring systems face challenges in maneuverability, scalability, and efficiency, making aerial solutions a compelling alternative. However, designing an effective control system for UAVs in a constrained greenhouse environment requires robust simulation frameworks to validate path planning, sensor integration, and plant inspection strategies before real-world deployment.

A simulation framework serves as a cost-effective and risk-free platform to test and optimize UAV control algorithms, enabling researchers to refine trajectory planning, obstacle avoidance, and sensor fusion techniques for real-time plant health assessment. This study presents a comprehensive simulation framework for quadrotor control, incorporating realistic environmental modeling, physics-based flight dynamics, and computer vision based plant monitoring techniques. By leveraging simulation, the proposed system facilitates precise way-point navigation, robust control, and imaging for crop analysis. The findings from this simulation framework will enhance the development of real-world quadrotor applications, contributing to precision agriculture and autonomous greenhouse management. In this research, the simulation environment is developed using ROS-I (Noetic) version on Ubuntu 20.04, alongside MATLAB 2024. The simulation is done on Intel(R) Core(TM) i9-14900KF 3.20 GHz processor desktop computer.

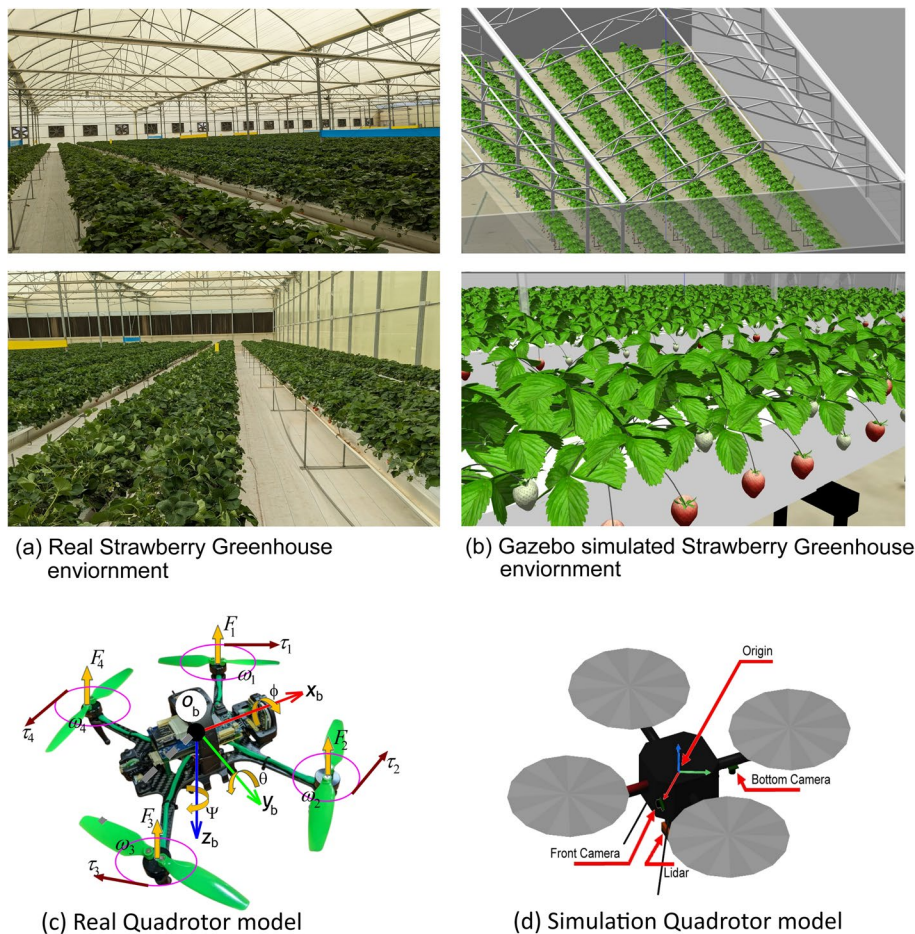


Fig. 17 Representation of a greenhouse strawberry farm, including greenhouse structure, strawberry fruits and leaves **a** a real-world environment, **b** a Gazebo-simulated environment, **c** Real quadrotor model, and **d** Simulation quadrotor model

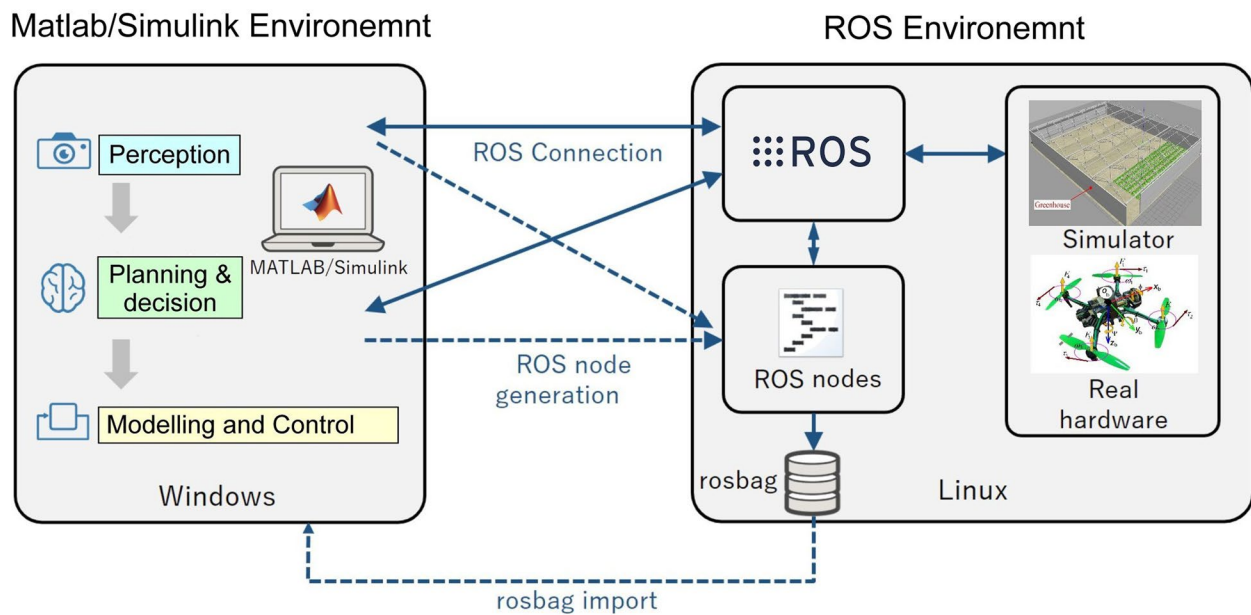


Fig. 18 Schematic diagram of the Matlab/Simulink-ROS communication network

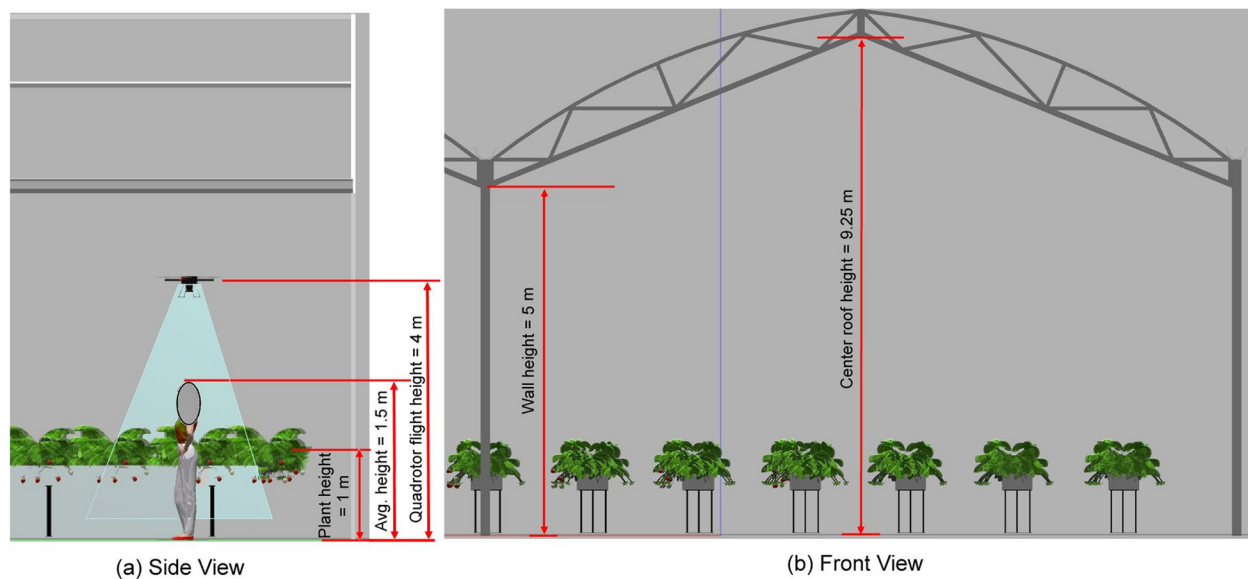


Fig. 19 Overall specification of the greenhouse design for quadrotor flight

Gazebo and ROS have been widely adopted in various robotics research projects for simulation and control purposes [51, 60]. Similarly, a detailed simulation study was conducted to analyze the dynamics and control of quadrotor UAVs [35]. ROS combine with computational tools such as bondgraph, matlab has been implemented [4, 59]. The greenhouse 3D model is developed within the Gazebo environment, as illustrated in Fig. 17a and b. This model serves as a digital twin representation of the Silal greenhouse in Abu Dhabi, accurately replicating the real-world strawberry crop greenhouse in a virtual simulation

framework. The environment includes objects such as greenhouse structure, terrain, stands, ripe strawberries, unripe strawberries, and leaves. Figure 17c represents the Gazebo model of the quadrotor for simulation environment. The quadrotor is equipped with front camera, bottom camera and bottom Lidar sensor for visualization of the objects.

Simulink-Gazebo co-simulation provides a powerful framework for designing, testing, and validating robotic control algorithms by integrating MATLAB/Simulink with the physics-based simulation capabilities of Gazebo.

This approach enables real-time simulation of robotic systems before deploying them on actual hardware. By utilizing ROS as the communication bridge, Simulink can subscribe to sensor data from Gazebo, process it in real time, and publish control commands back to the simulated robot as depicted in Fig. 18.

Predefined way-points serve as inputs to the controller, guiding the quadrotor to follow the generated trajectory. The primary trajectory is designed specifically for crop monitoring applications within a greenhouse using a quadrotor. The overall greenhouse design specifications, including plant height, average operator height, quadrotor flight height, sidewall height, and center roof height, are illustrated in Fig. 19. As shown in Fig. 19, the quadrotor follows a trajectory at an approximate height of 4 meters from ground for selected strawberry crop. In this research, it is assumed that there are no obstacles in the quadrotor's path at that height.

The quadrotor operates at a speed of 2 m/s during its monitoring operation. In this study the Modal AI starling2 quadrotor UAV is used for simulation modeling and real-world testing.

Results

Figure 20 presents the desired trajectory, the MATLAB-based actual path, and the controller's performance as followed by the quadrotor. As shown in Fig. 20b, the translational and rotational trajectories of the quadrotor closely follow the desired reference model response based on predefined way-points. Furthermore, the 3D trajectory tracking in Fig. 20a demonstrates that the controller ensures accurate tracking of the desired reference

model response by stabilizing the quadrotor as it flies over the strawberry crop in the greenhouse environment.

The proposed deep learning based detection model and quadrotor control algorithm are validated in a Gazebo-based simulated greenhouse environment and actual greenhouse environment. The quadrotor navigates through the strawberry greenhouse, following a predefined way-point trajectory to ensure accurate and controlled movement. Figures 21 and 22 shows the simulation and real quadrotor controller testing and crop monitoring application.

Figure 21a illustrates the greenhouse environment as the quadrotor ascends to the desired altitude during the initial phase. The bottom and front camera views captured by the quadrotor are presented in Fig. 21b.1 and b.2. Additionally, Fig. 21c and d showcase the quadrotor's position throughout the trajectory tracking process. The proposed trajectory tracking controller was evaluated at the Silal greenhouse facility in Abu Dhabi, UAE, using a square trajectory as a test case. As illustrated in Fig. 22, the quadrotor successfully navigated the greenhouse environment by following the predefined trajectory through designated way-points.

Figure 22a and b illustrate the quadrotor UAV operating within a real-world greenhouse environment at Silal Farm in Abu Dhabi, UAE. Figure 22c presents the view captured by the quadrotor's front camera, while Fig. 22d shows the Modal AI Starling 2 quadrotor's web interface displaying IMU and Gyro sensor data during trajectory tracking along predefined waypoints. As demonstrated in Fig. 22d, the controller achieves satisfactory performance for the intended trajectory tracking (path in cyan color).

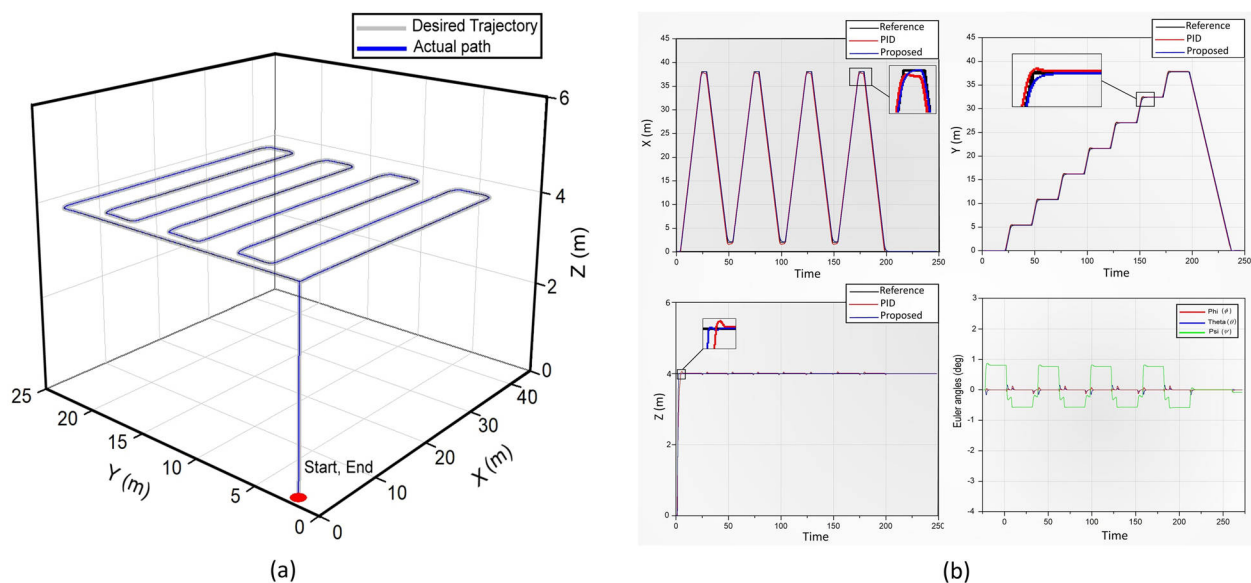


Fig. 20 Trajectory tracking performance of the quadrotor controller

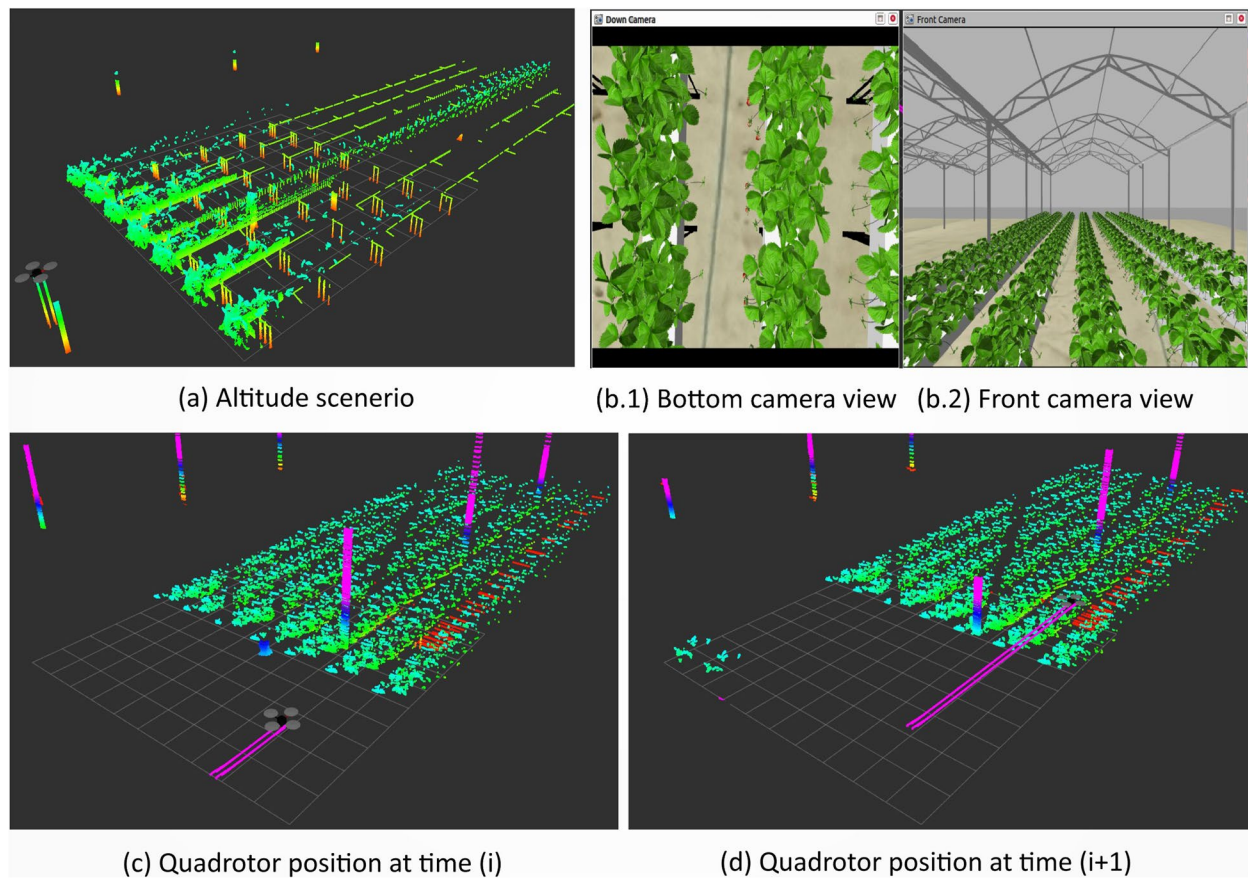


Fig. 21 Visualization of the quadrotor's movement and camera perspectives in the greenhouse environment. **a** Altitude scenario showing the quadrotor reaching the desired height. **b.1** Bottom camera view capturing the crops below. **b.2** Front camera view displaying the greenhouse layout. **c** Quadrotor position at time i during trajectory tracking. **d** Quadrotor position at time $i+1$ indicating its progression along the predefined path

For the crop monitoring application, specifically for counting ripe strawberries, the trained model is tested on video footage captured by the quadrotor during navigation both in simulation and real-time. The trained weights (best.pt) are integrated into the ROS interface to detect and track ripe strawberries in real-time. A line counter (solid red line) method is employed to count the detected ripe strawberries throughout the monitoring process. The results, presented in Fig. 23, illustrate the detection process from the initial scanning stage to the final stage. As shown in Fig. 23, the detection and counting algorithm effectively identifies and tracks ripe strawberries and provide separate id's to each detected and classified strawberry during flight operations. Table 5 presents the strawberry counting results compared to the ground truth data. The counting accuracy (%) for ripe strawberries in both simulation and real-world greenhouse environments is satisfactory, indicating the method's effectiveness and its potential for further enhancement in future research.

Discussion

The achieved detection performance metrics of the integrated MSD-quadrotor system demonstrate significant potential for transforming greenhouse agricultural operations, with the YOLOv9-GELAN model's exceptional precision (1.00 at confidence 0.943) translating directly into labor reduction opportunities in commercial greenhouse settings. Traditional manual strawberry monitoring requires skilled workers to inspect approximately 200-300 plants per hour, whereas the proposed system can systematically scan equivalent areas in 15-20 minutes with consistent accuracy, representing a 75-80% reduction in monitoring time and associated labor costs. The model's ability to distinguish between ripe and unripe strawberries with 99.4% mAP@0.5 enables precise harvest timing decisions critical for maximizing fruit quality and market value.

The hybrid PID-LQR controller's trajectory tracking accuracy (± 5 cm positioning precision) enables seamless integration with existing greenhouse automation

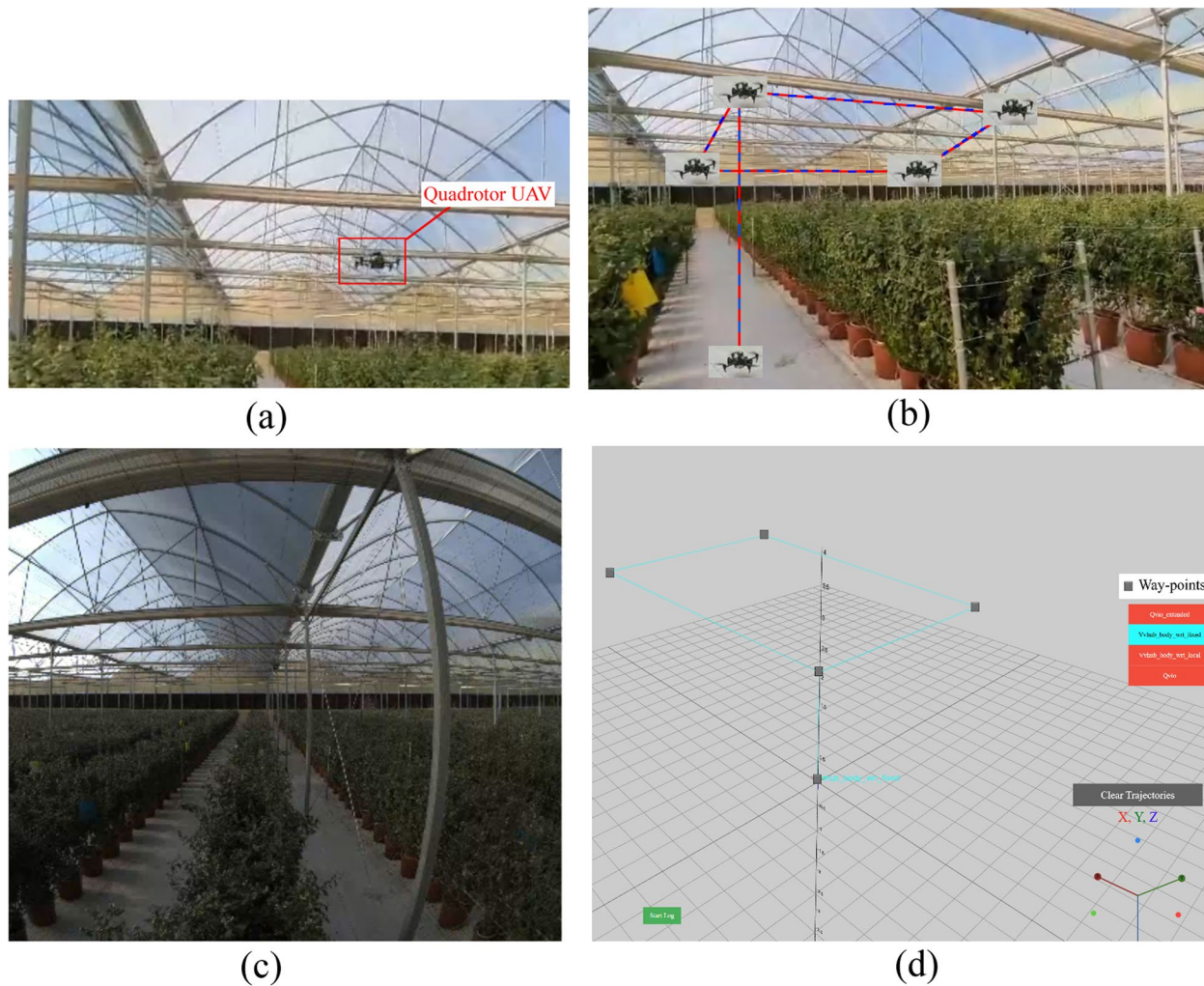


Fig. 22 Real-time validation of the proposed hybrid PID-LQR trajectory tracking controller in greenhouse environment: **a** quadrotor UAV operation, **b** planned trajectory execution with waypoint navigation, **c** onboard camera perspective during flight, and **d** IMU/gyroscope sensor data visualization showing precise waypoint tracking and flight path monitoring in simulation interface

infrastructure, with the system's Python-based implementation facilitating compatibility with standard greenhouse management platforms. The demonstrated processing speeds of 15–20 minutes per greenhouse section position the system as commercially viable for medium to large-scale operations.

While the system demonstrates excellent performance in controlled greenhouse environments, practical deployment considerations include the hybrid controller's complexity requiring specialized maintenance knowledge, the 8–12% reduction in flight time due to computational overhead limiting continuous monitoring capabilities, and potential regulatory compliance requirements for autonomous UAV operations in agricultural settings. The future research roadmap's focus on IoT platform integration addresses critical needs for comprehensive farm management systems, where integration with existing greenhouse sensors will enable correlation analysis

between environmental conditions and fruit development patterns, supporting predictive modeling for yield forecasting and quality optimization. The planned model optimization through parameter reduction techniques will facilitate deployment on edge devices, reducing dependence on high-performance computing hardware and enabling cost-effective scaling for smaller agricultural operations, positioning the system as a foundational component for next-generation precision agriculture implementations in controlled environment agriculture.

Conclusion

This research introduces an advanced deep learning framework for identifying ripe strawberries in greenhouse settings through quadrotor-based systems. The study implements a YOLOv9-GELAN algorithm on the aerial platform, demonstrating superior performance compared to existing methodologies with achieved metrics of

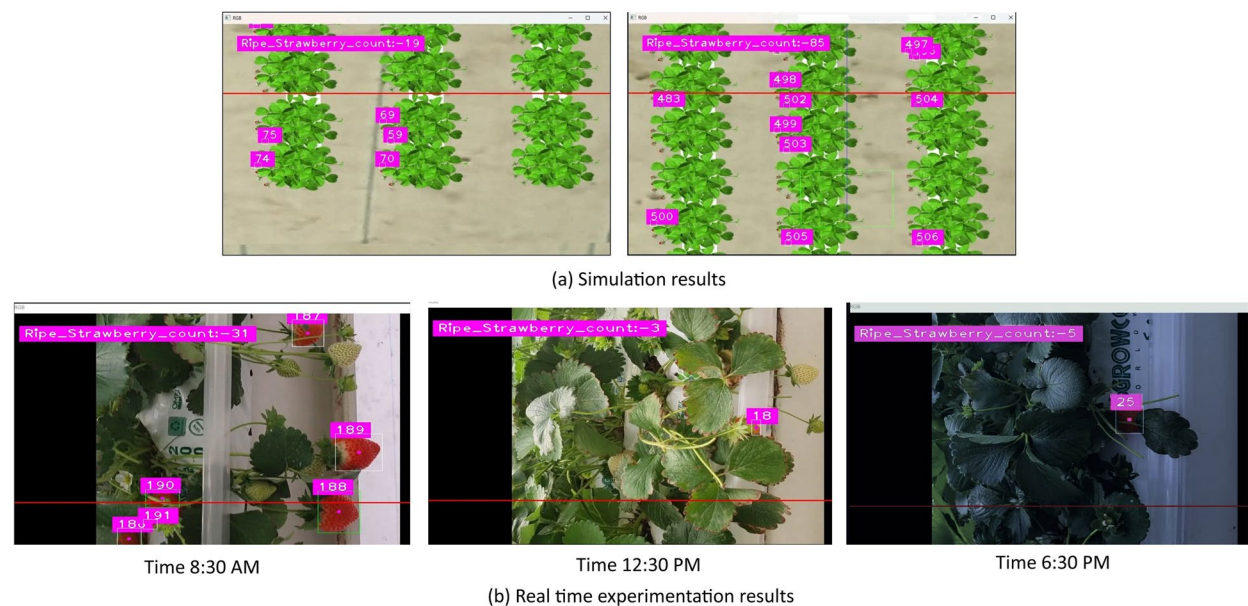


Fig. 23 Real time ripe strawberries detection and counting during crop monitoring task by quadrotor in simulation and real-time experimentation at different time frames

Table 5 Counting accuracy of ripe strawberries in real-world experiment

	Real-world green-house environment experiment		
Avg. ripe strawberry counting	Drone video	Ground truth	Accuracy (%)
(8:30 AM)	45	47	95.7
(12:30 PM)	46	49	93.8
(6:30 PM)	44	49	89.7

99.10% precision, 99.0% recall, and real-time processing at 23 fps. A comprehensive dataset incorporating both synthetic and real strawberry imagery was developed to train the deep learning architecture. Model validation was conducted across simulated and actual greenhouse environments to ensure robust performance. Additionally, a hybrid trajectory control system combining PID and LQR techniques was developed and implemented for quadrotor navigation, surpassing conventional control approaches in both computational and experimental evaluations. The complete system exhibited strong practical utility for strawberry detection and enumeration tasks, delivering 94.6% accuracy in simulated conditions and 95.7% accuracy during field deployment. Prospective work will focus on incorporating sophisticated perception technologies such as visual odometry to enhance navigation precision, alongside integrating supplementary sensor systems to improve environmental perception and advance autonomous monitoring capabilities.

Acknowledgements
This research is supported by Khalifa University Center for Robotics and Autonomous Systems (KUCARS), under Grant No. RCI-2018-KUCARS-T4.

Institutional review board
Not applicable.

Clinical trial number
Not applicable.

Informed consent
Not applicable.

Authors' contributions
R.S: As the first author, responsible for the main aspects of the research, including experimental design, data collection, analysis and interpretation, as well as drafting the initial manuscript. Conceptualization, methodology, software, writing-original draft, formal analysis, writing-reviews and editing. A.G: Implementation and result analysis, Reviewing and Editing the draft. I.H: As the principal corresponding author, responsible for the overall guidance of the research, team coordination, resources, final review, and correspondence with the journal.

Funding
Not applicable.

Data availability
The datasets used during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study did not involve human participants or animals, thus no ethics approval or consent to participate was required.

Competing interest
The authors declare no competing interests.

Received: 30 May 2025 / Accepted: 18 August 2025
Published online: 13 October 2025

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